

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

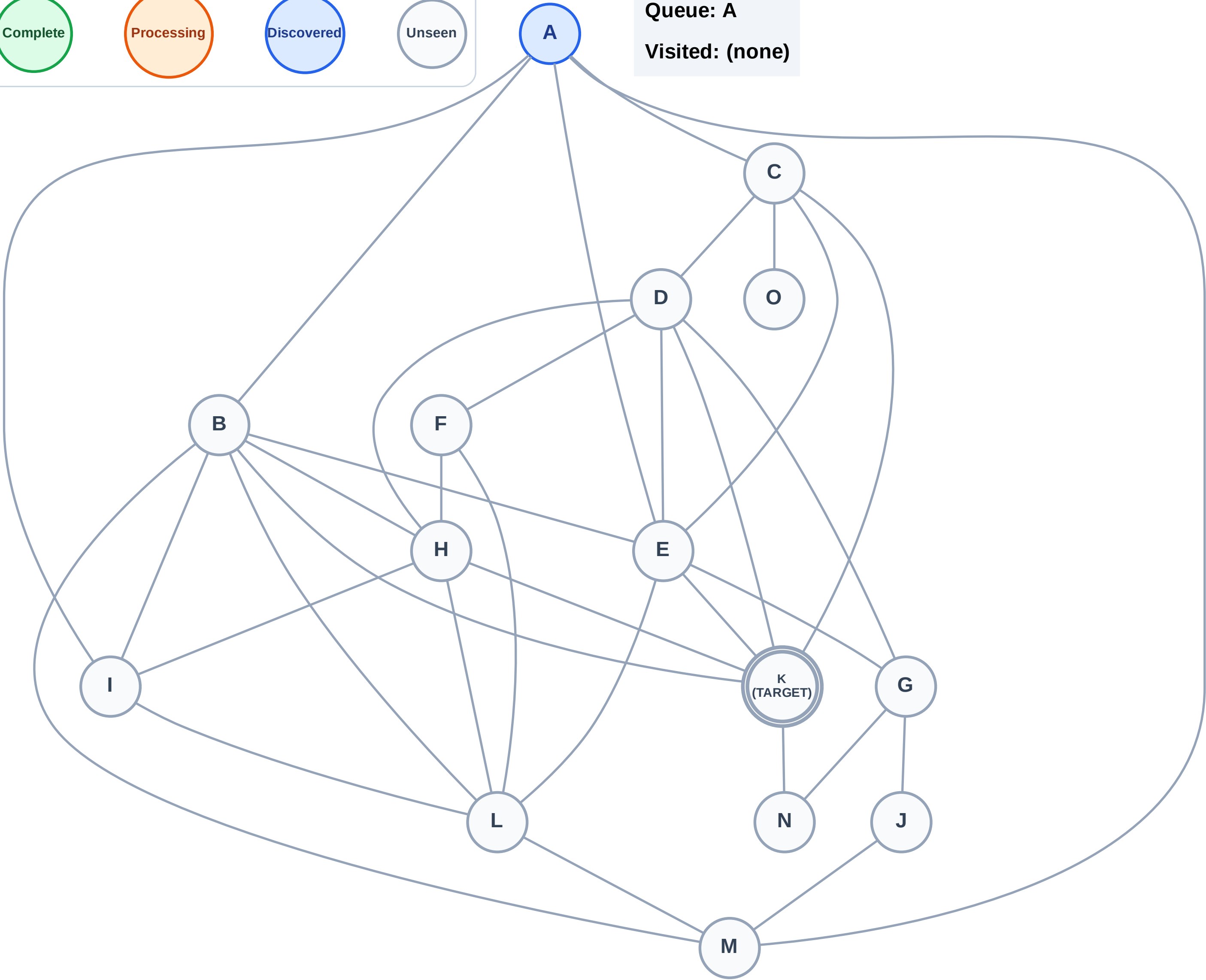
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

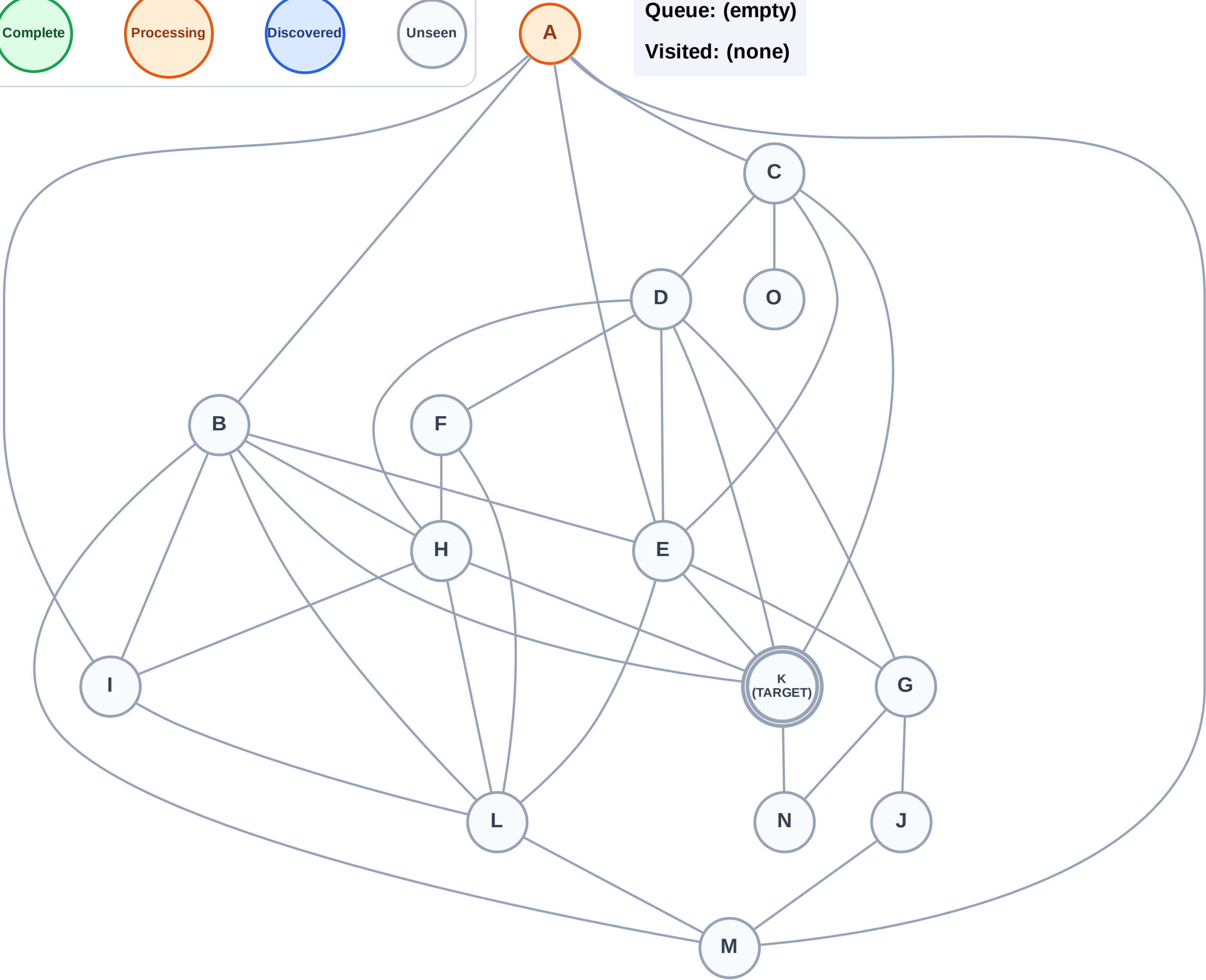
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, I, M from A

Legend

Complete

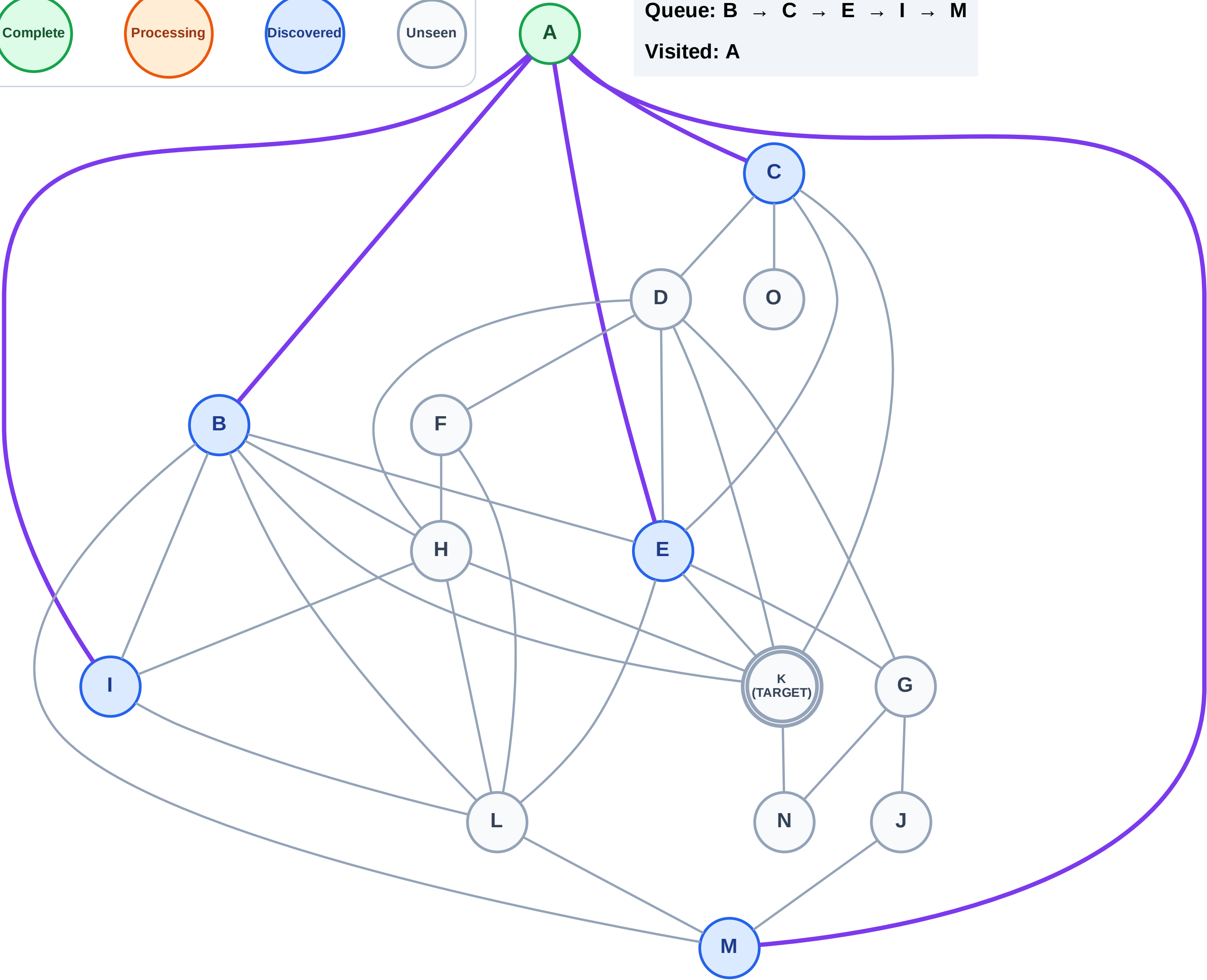
Processing

Discovered

Unseen

Queue: B → C → E → I → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

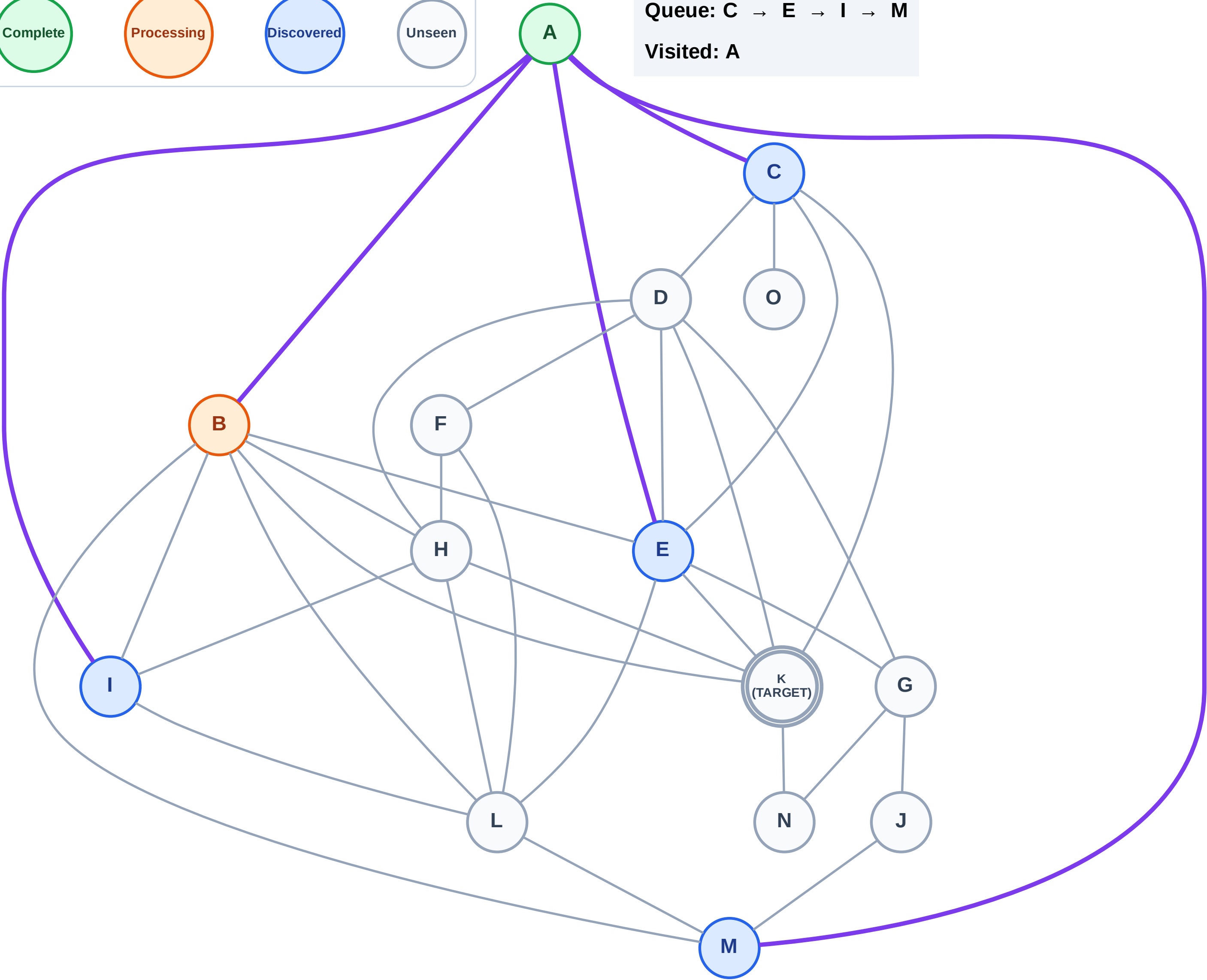
Processing

Discovered

Unseen

Queue: C → E → I → M

Visited: A



BFS Traversal

Step 5: Discover H, K, L from B

Legend

Complete

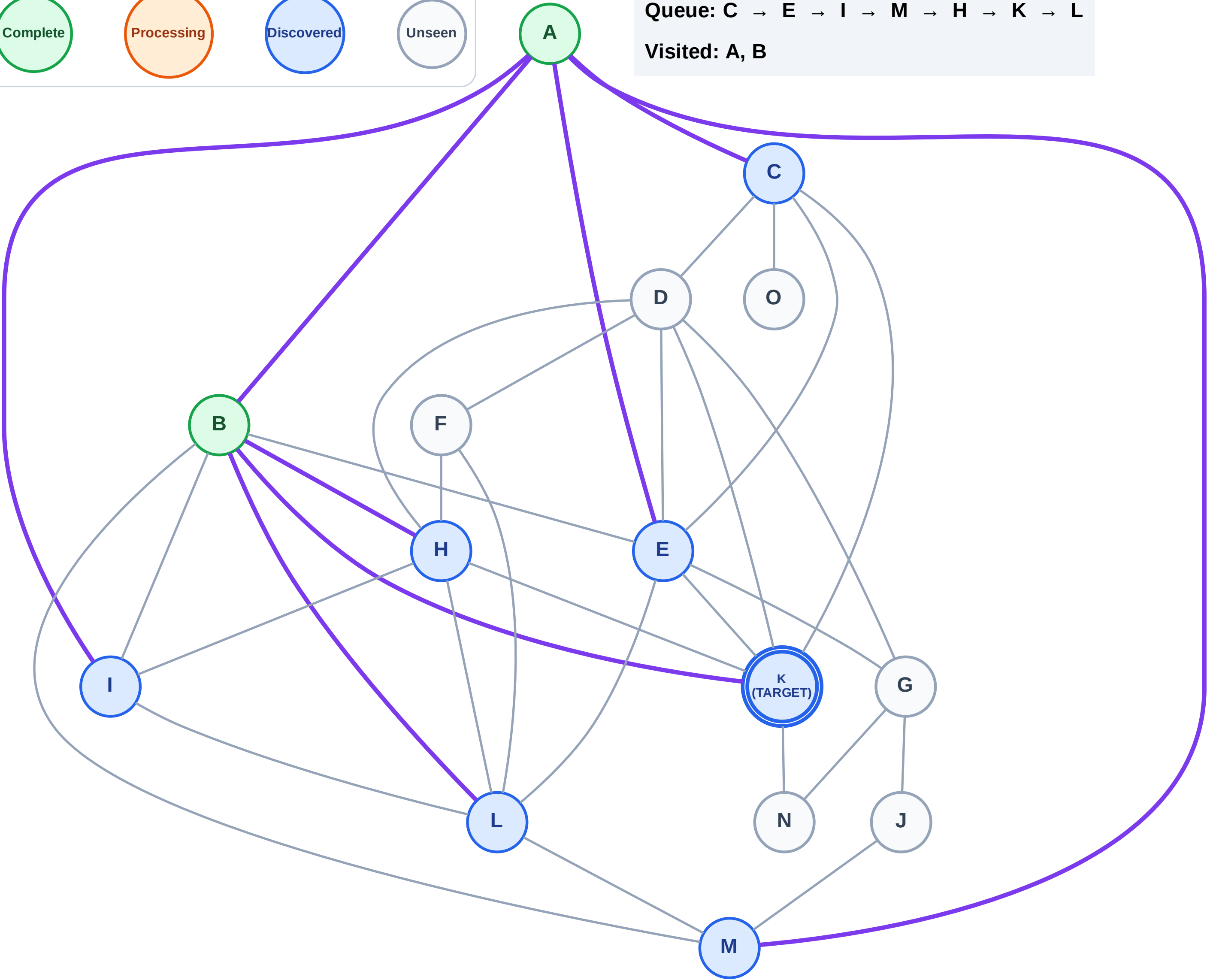
Processing

Discovered

Unseen

Queue: C → E → I → M → H → K → L

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

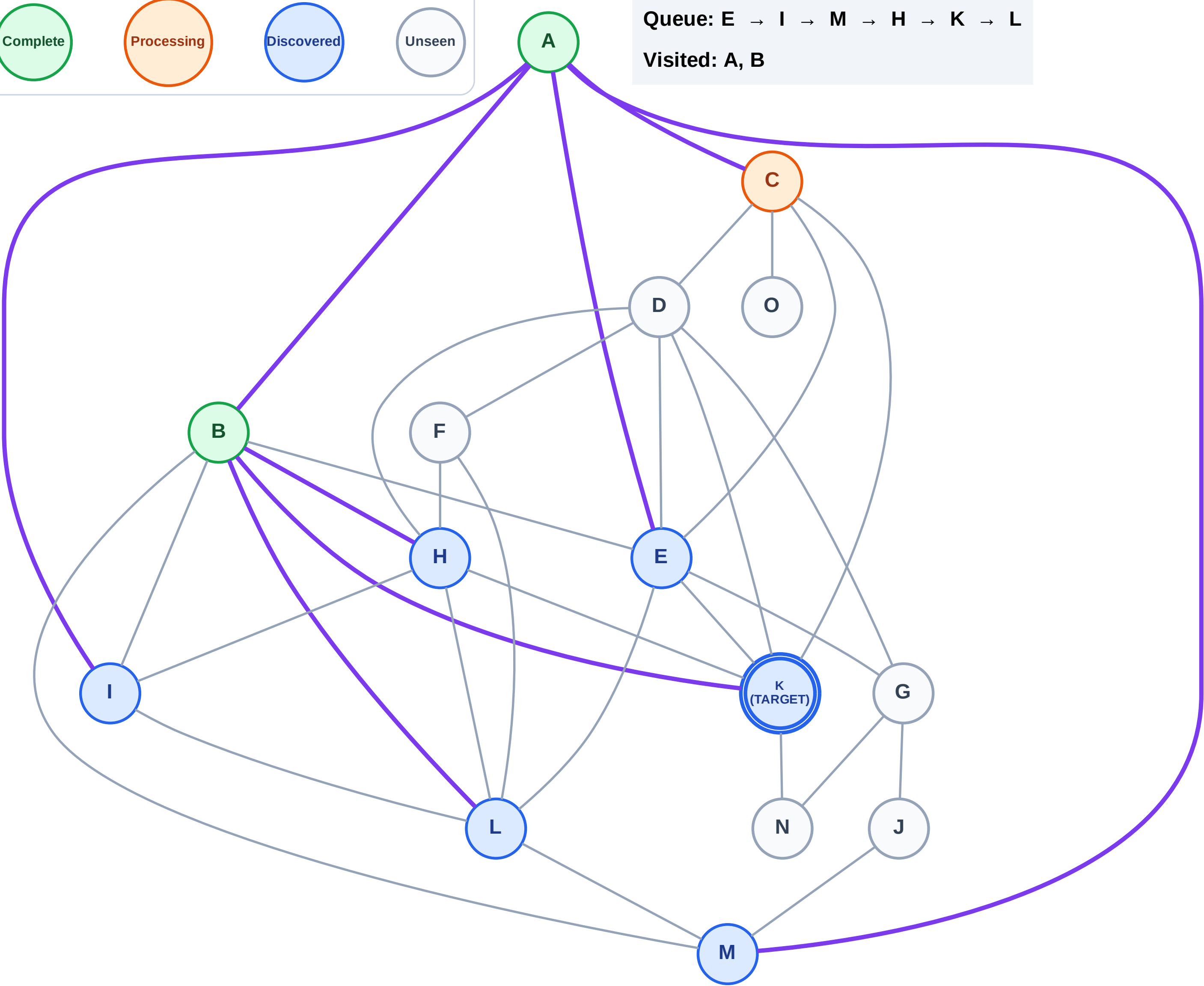
Processing

Discovered

Unseen

Queue: E → I → M → H → K → L

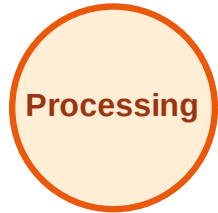
Visited: A, B



BFS Traversal

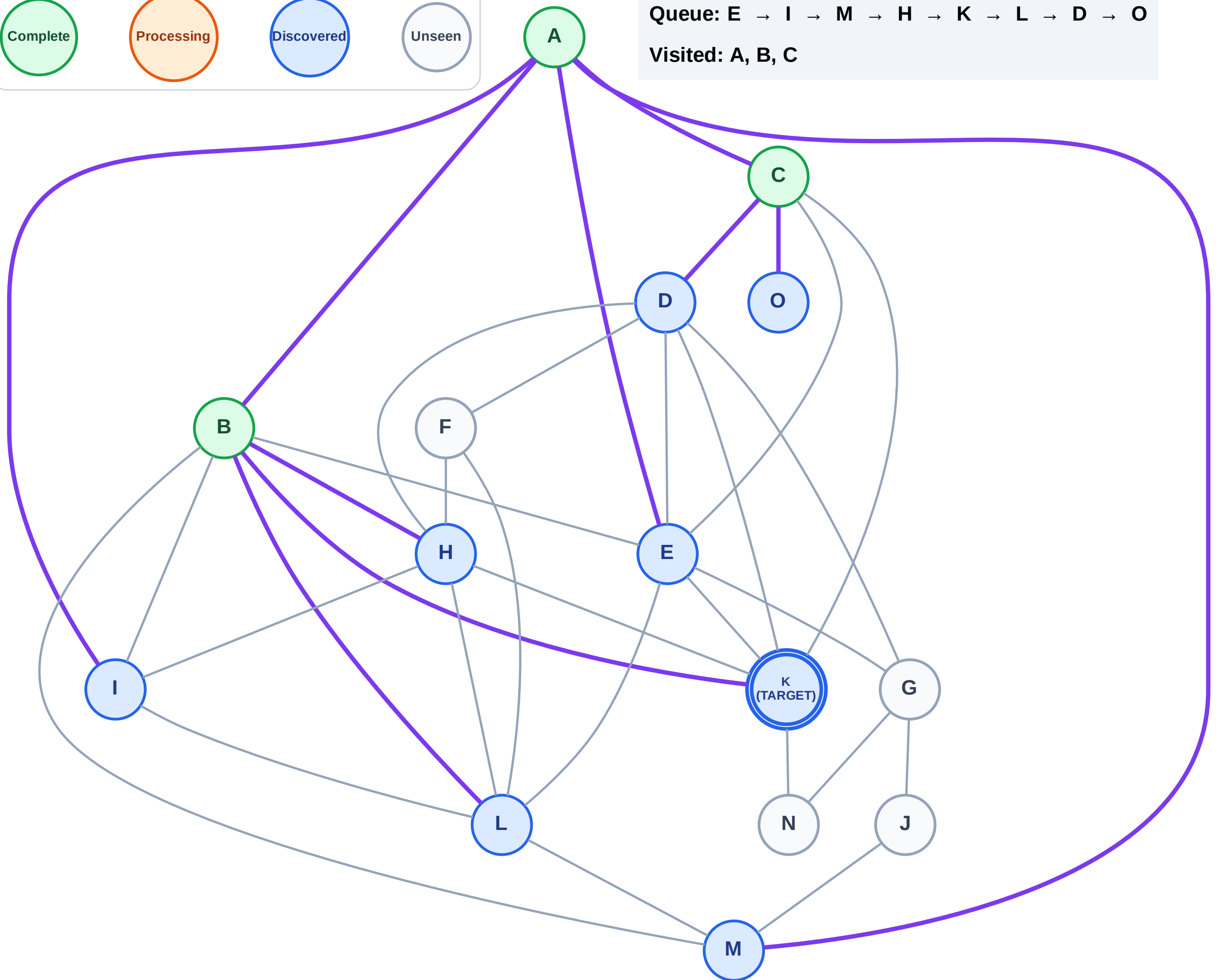
Step 7: Discover D, O from C

Legend



Queue: E → I → M → H → K → L → D → O

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

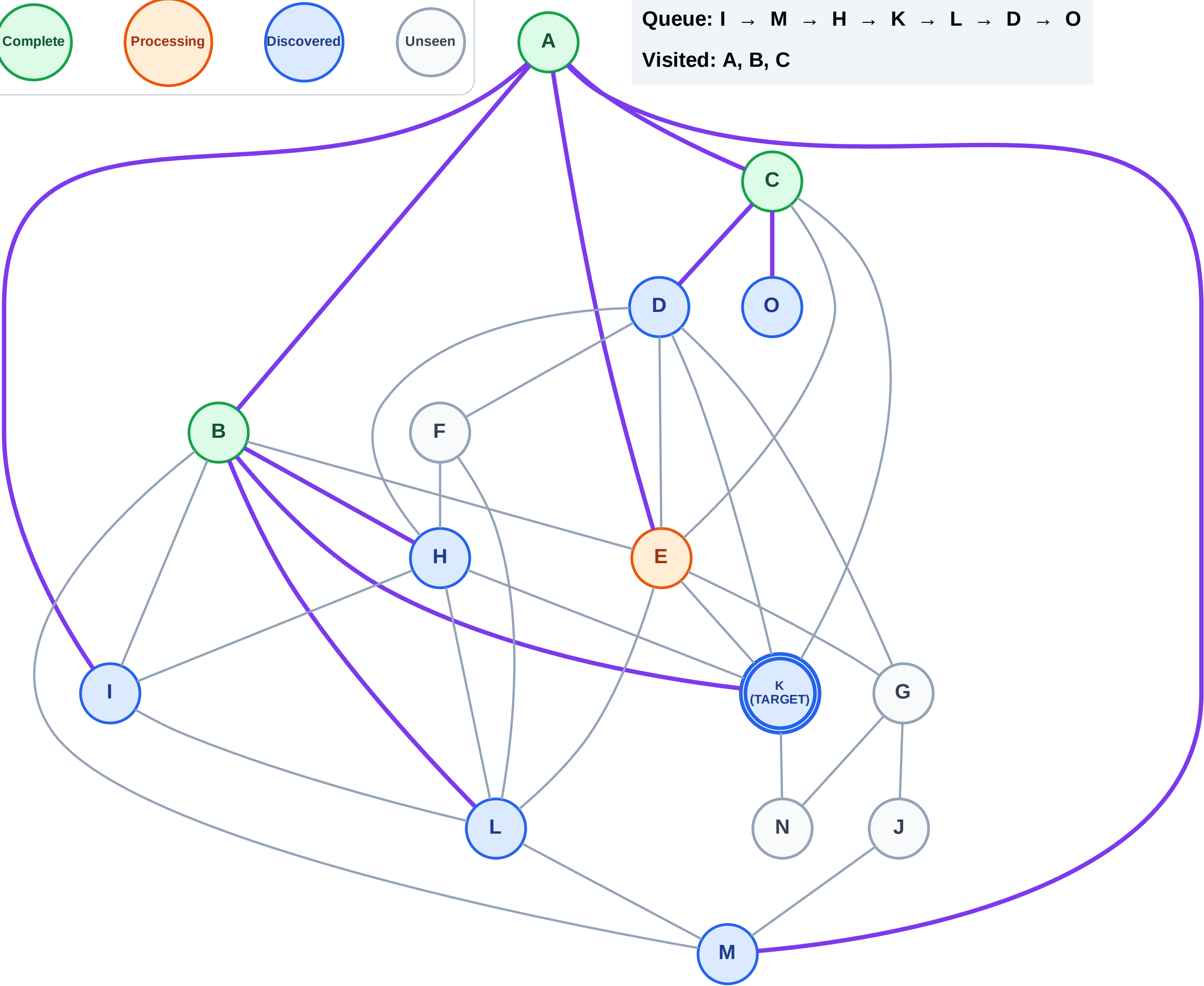
Processing

Discovered

Unseen

Queue: I → M → H → K → L → D → O

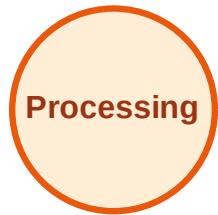
Visited: A, B, C



BFS Traversal

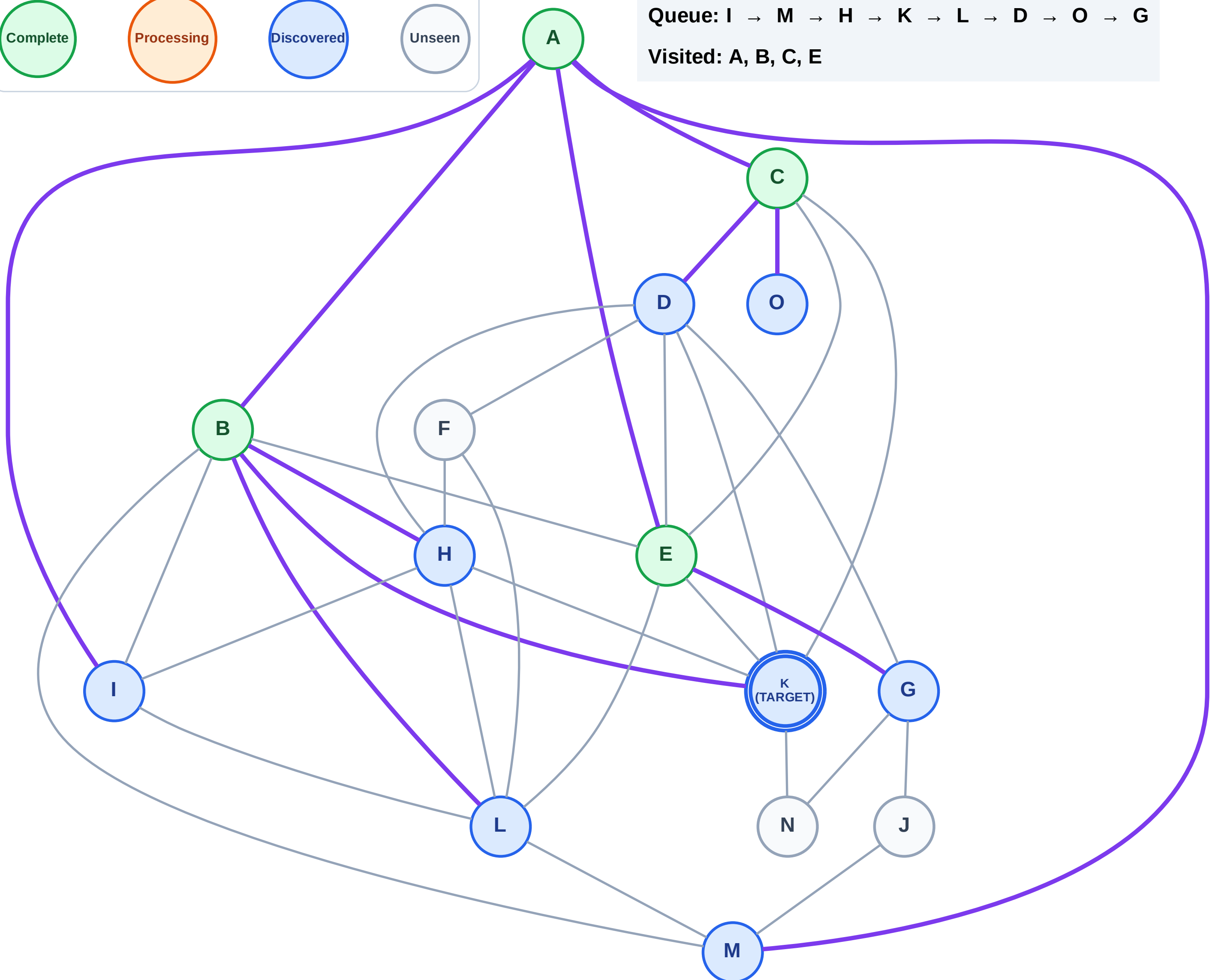
Step 9: Discover G from E

Legend



Queue: I → M → H → K → L → D → O → G

Visited: A, B, C, E



BFS Traversal

Step 10: Process node I

Legend

Complete

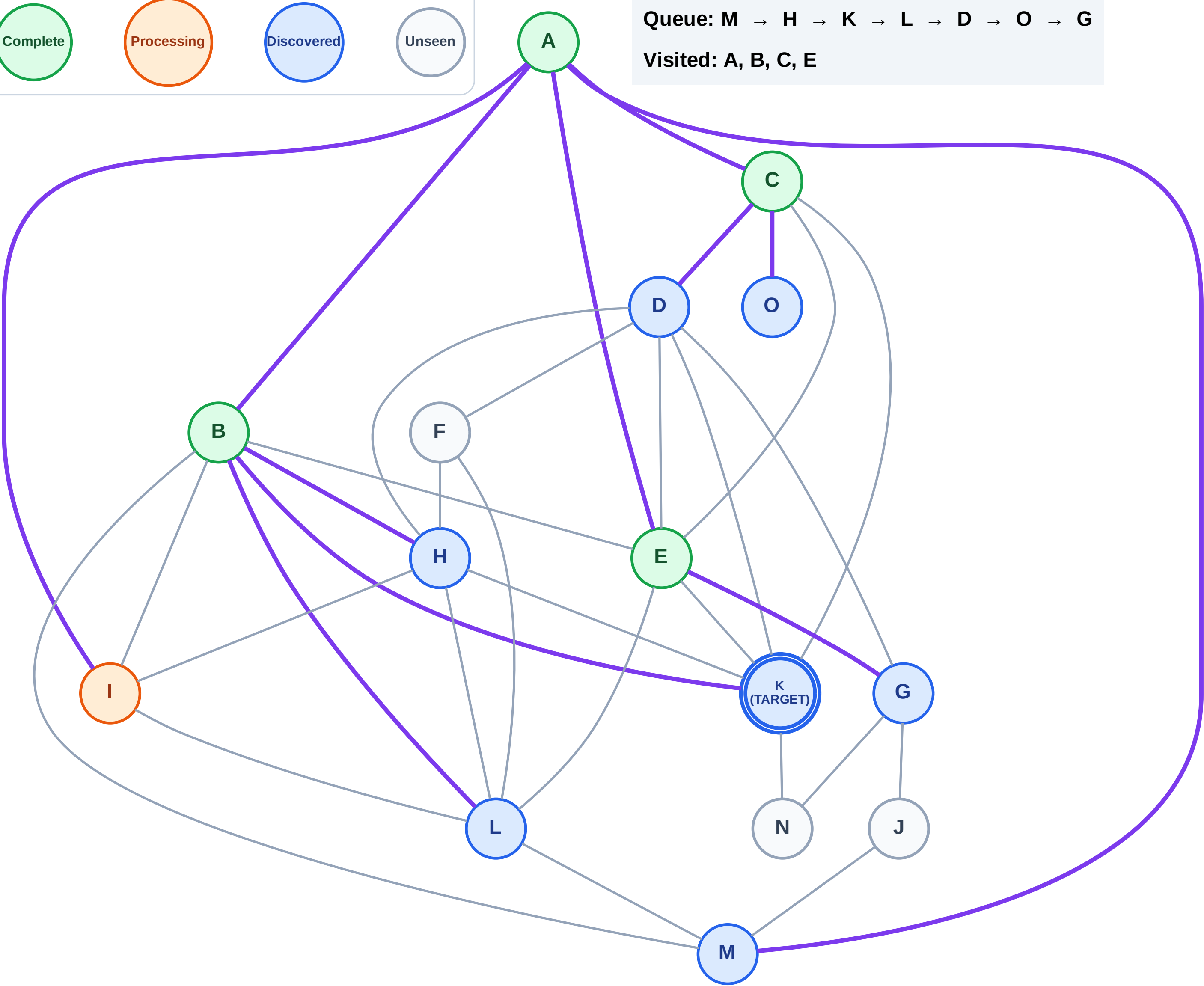
Processing

Discovered

Unseen

Queue: M → H → K → L → D → O → G

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

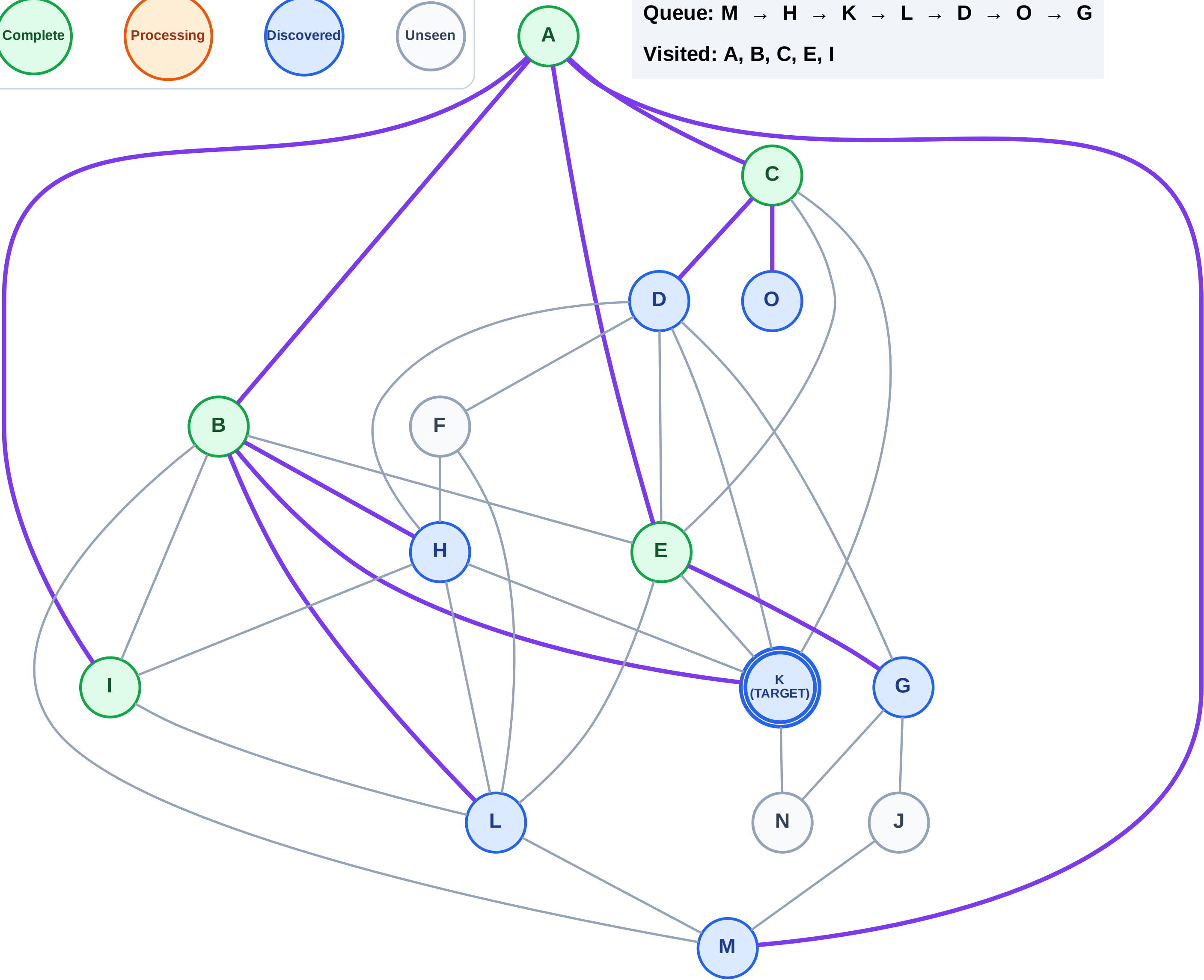
Processing

Discovered

Unseen

Queue: M → H → K → L → D → O → G

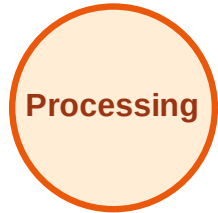
Visited: A, B, C, E, I



BFS Traversal

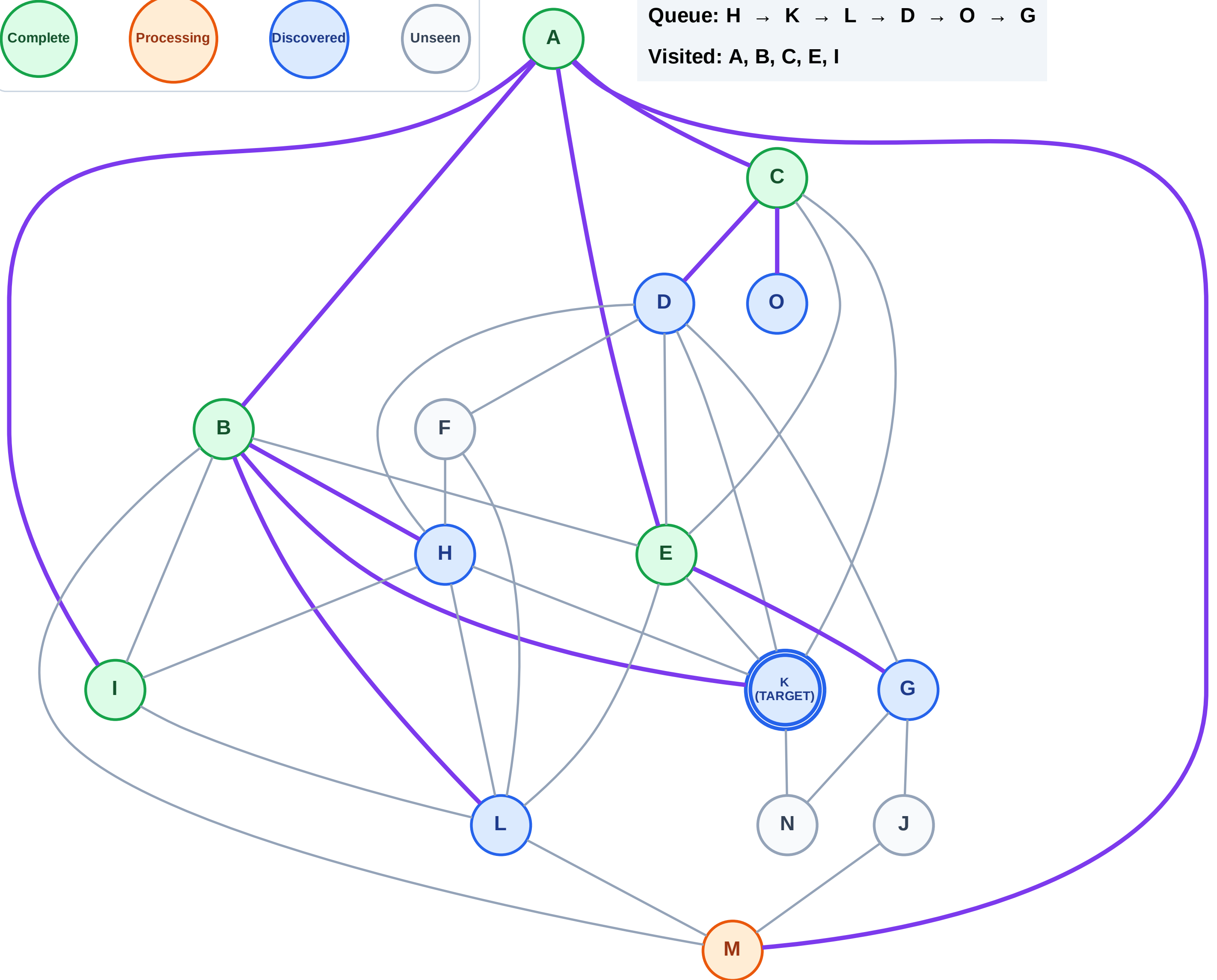
Step 12: Process node M

Legend



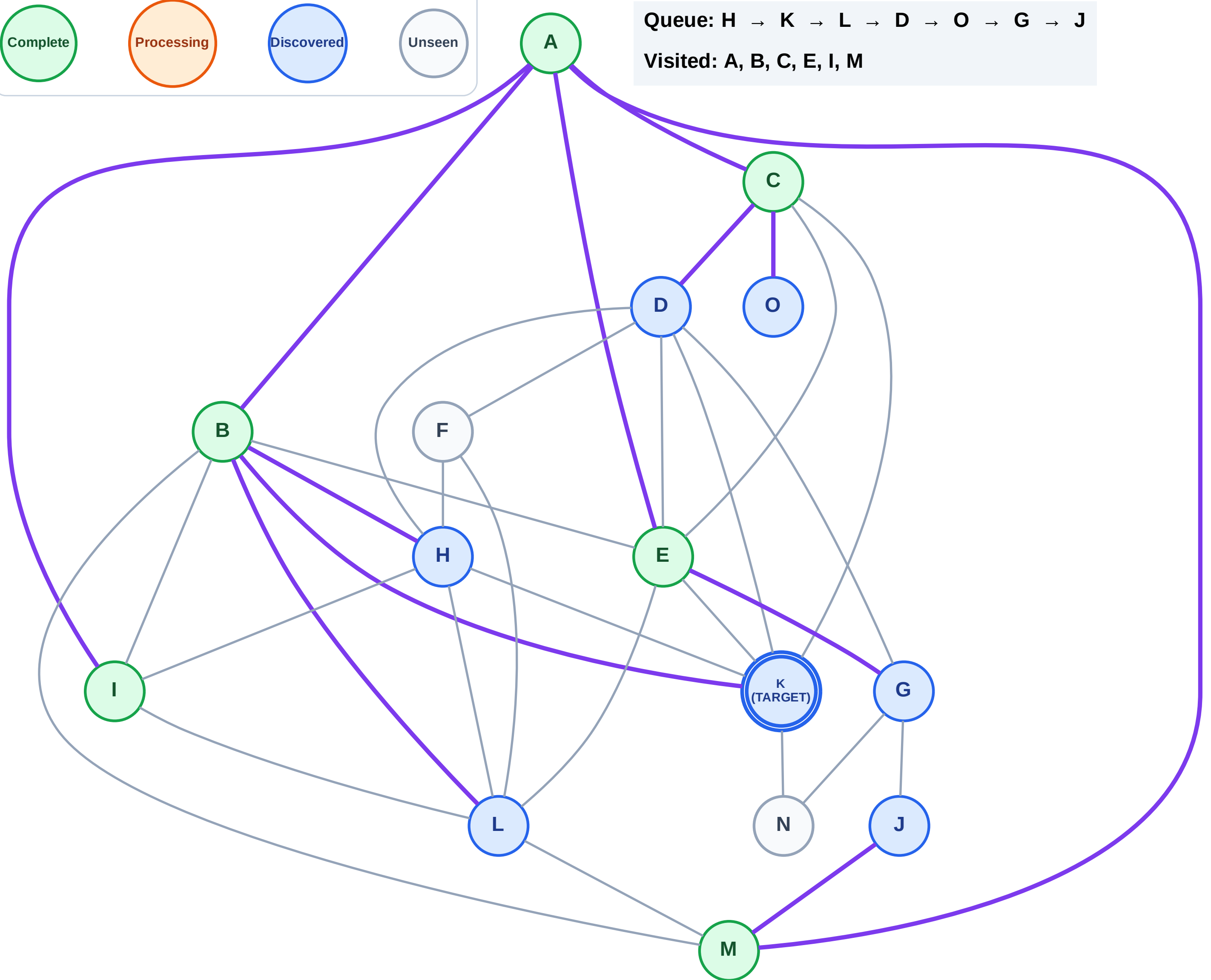
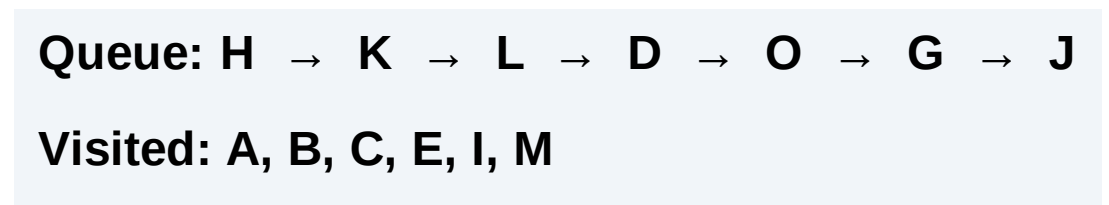
Queue: H → K → L → D → O → G

Visited: A, B, C, E, I



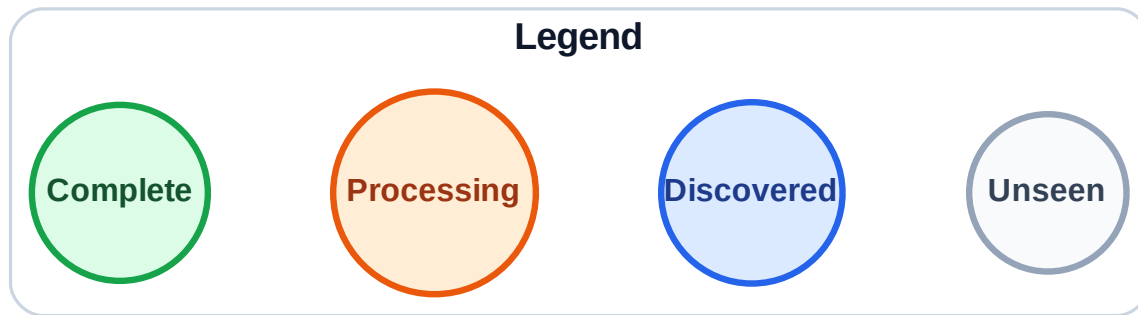
Step 13: Discover J from M

Step 13: Discover J from M



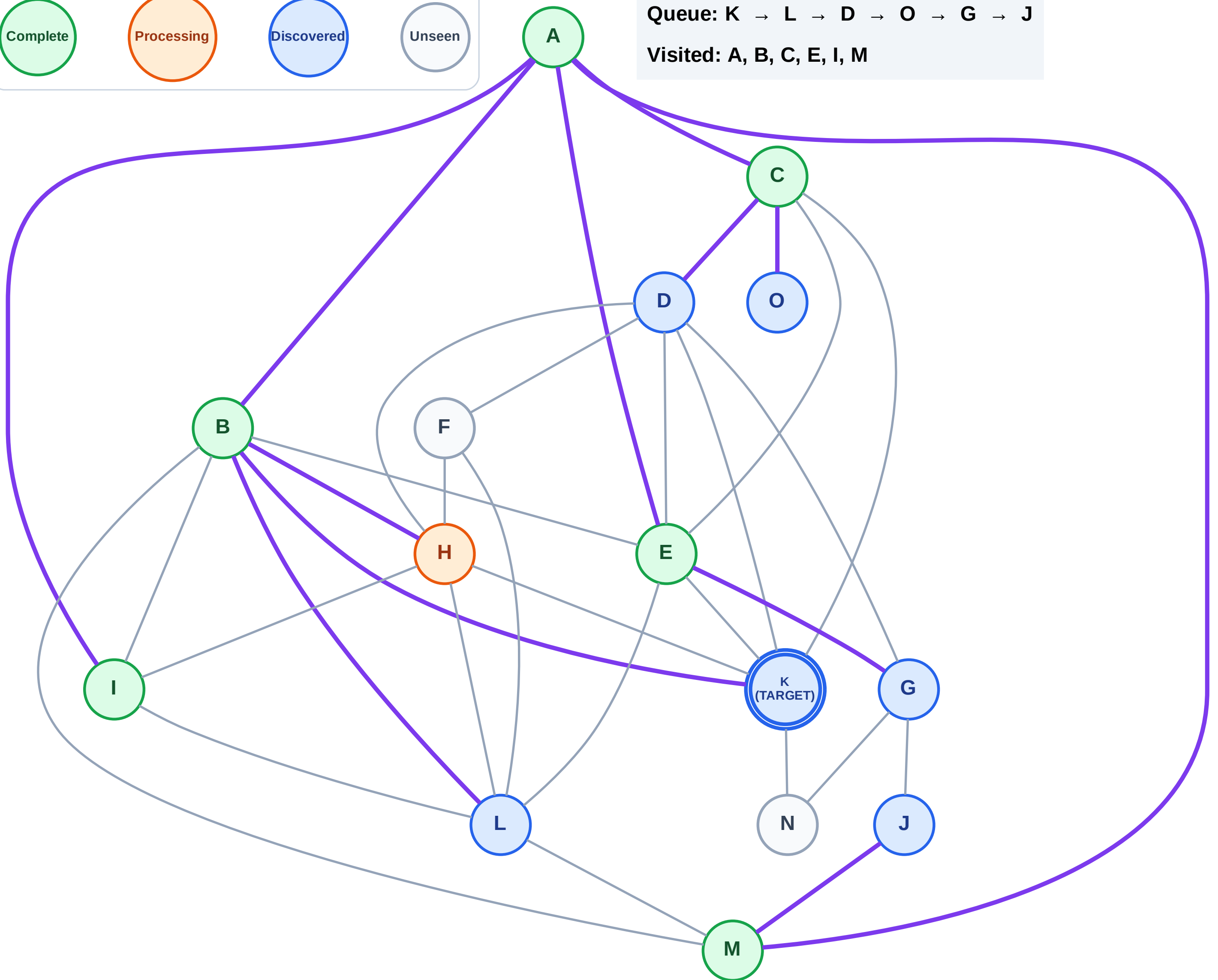
BFS Traversal

Step 14: Process node H



Queue: K → L → D → O → G → J

Visited: A, B, C, E, I, M



BFS Traversal

Step 15: Discover F from H

Legend

Complete

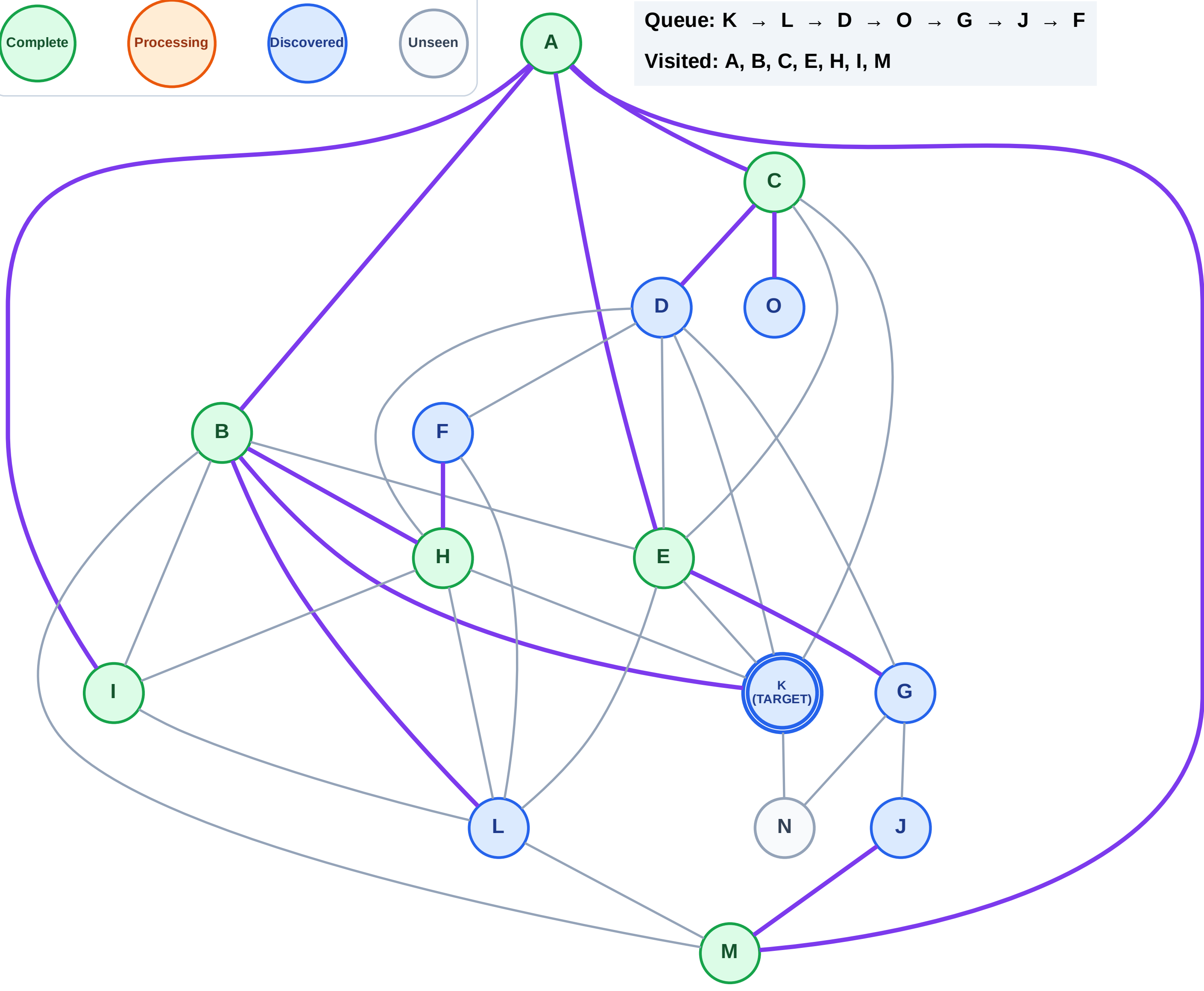
Processing

Discovered

Unseen

Queue: K → L → D → O → G → J → F

Visited: A, B, C, E, H, I, M



BFS Traversal

Step 16: Process node K

Legend

Complete

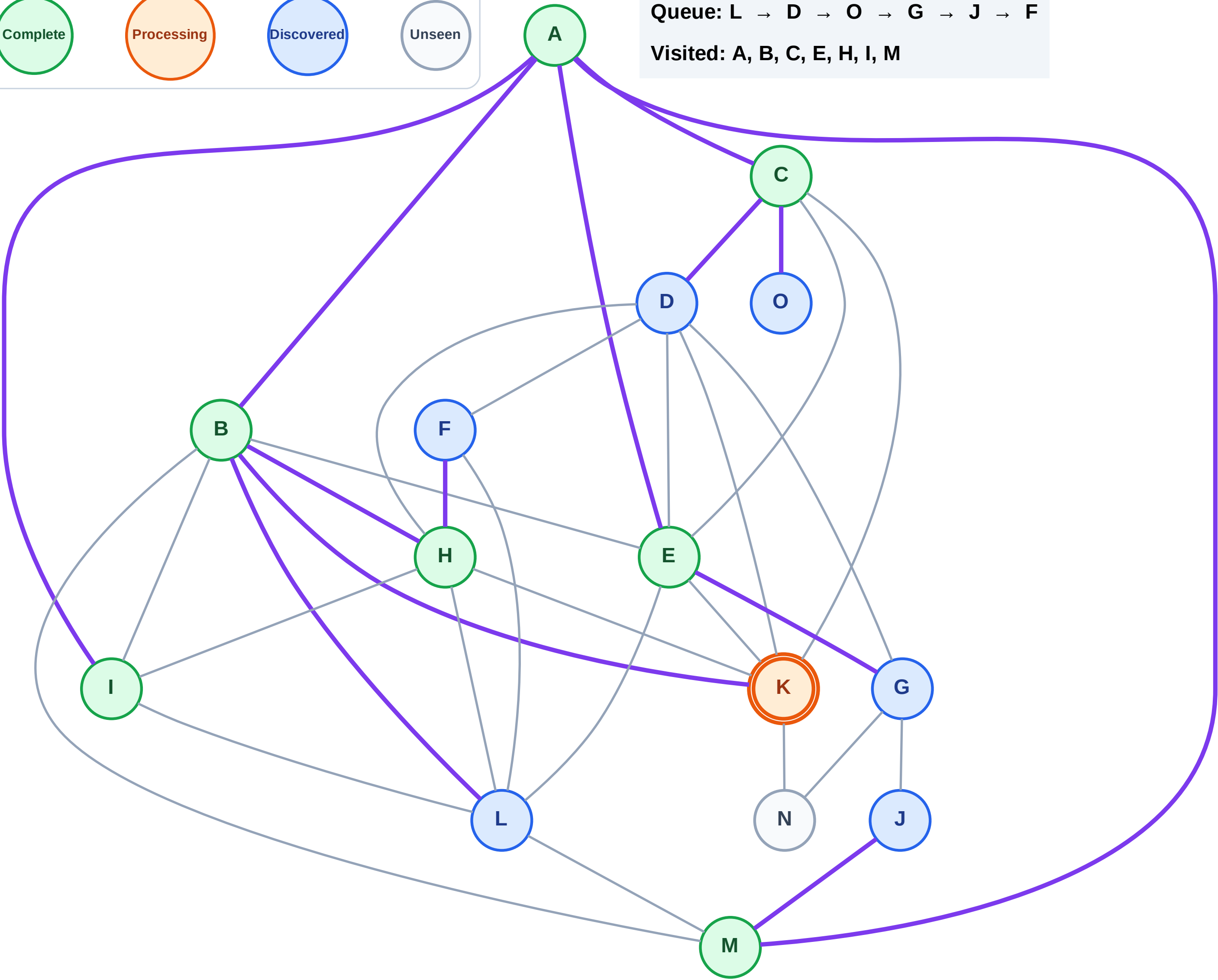
Processing

Discovered

Unseen

Queue: L → D → O → G → J → F

Visited: A, B, C, E, H, I, M



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

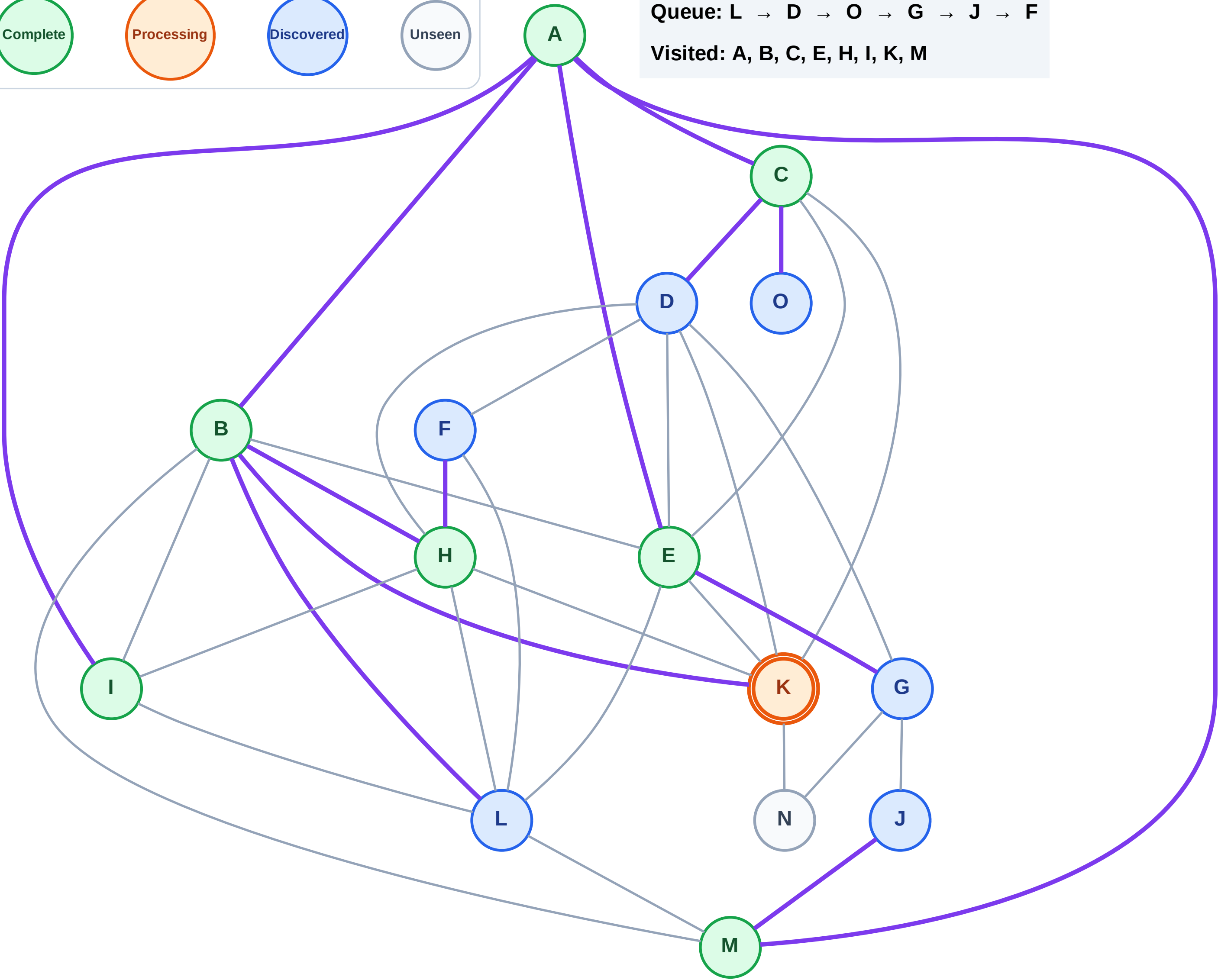
Processing

Discovered

Unseen

Queue: L → D → O → G → J → F

Visited: A, B, C, E, H, I, K, M



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

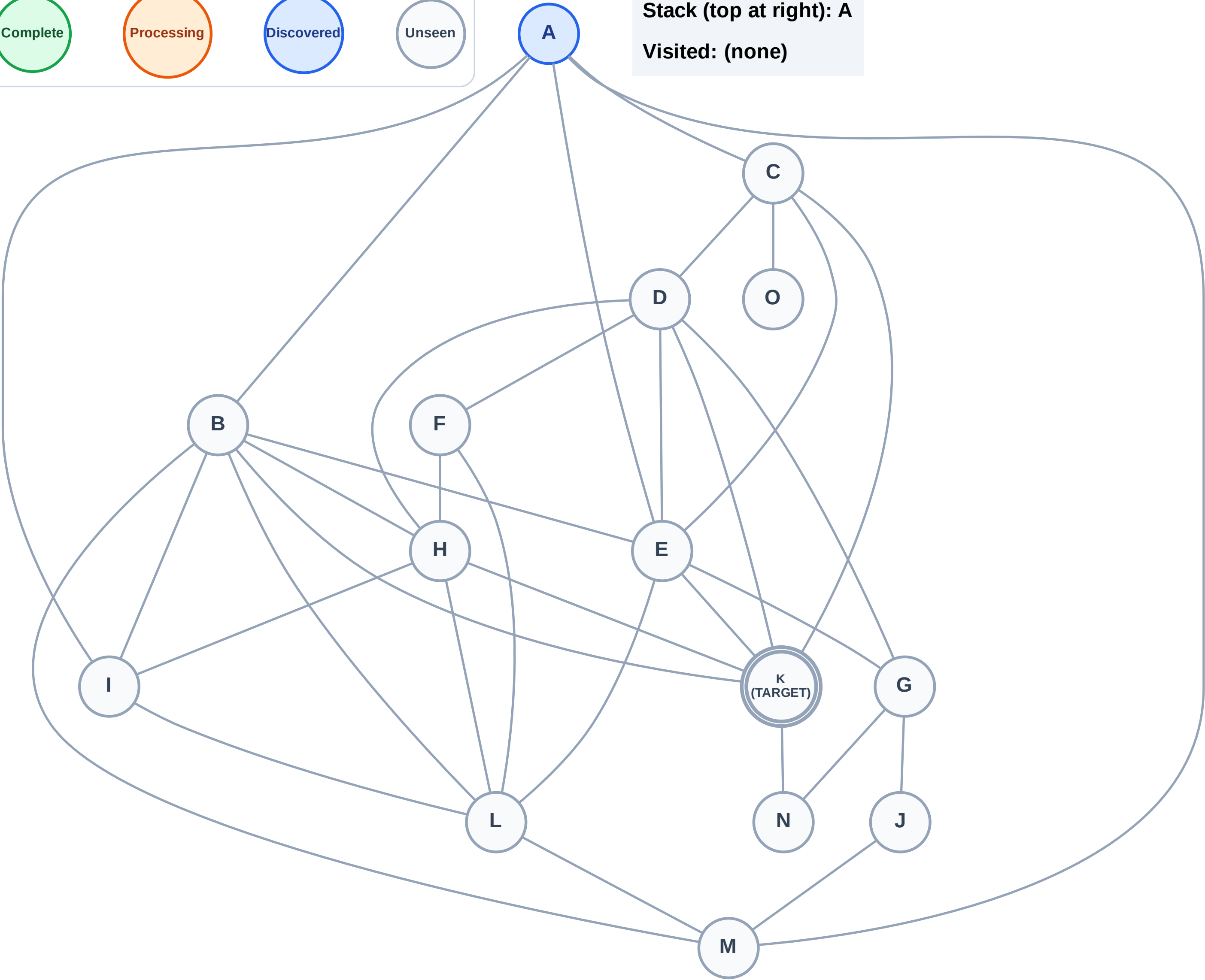
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

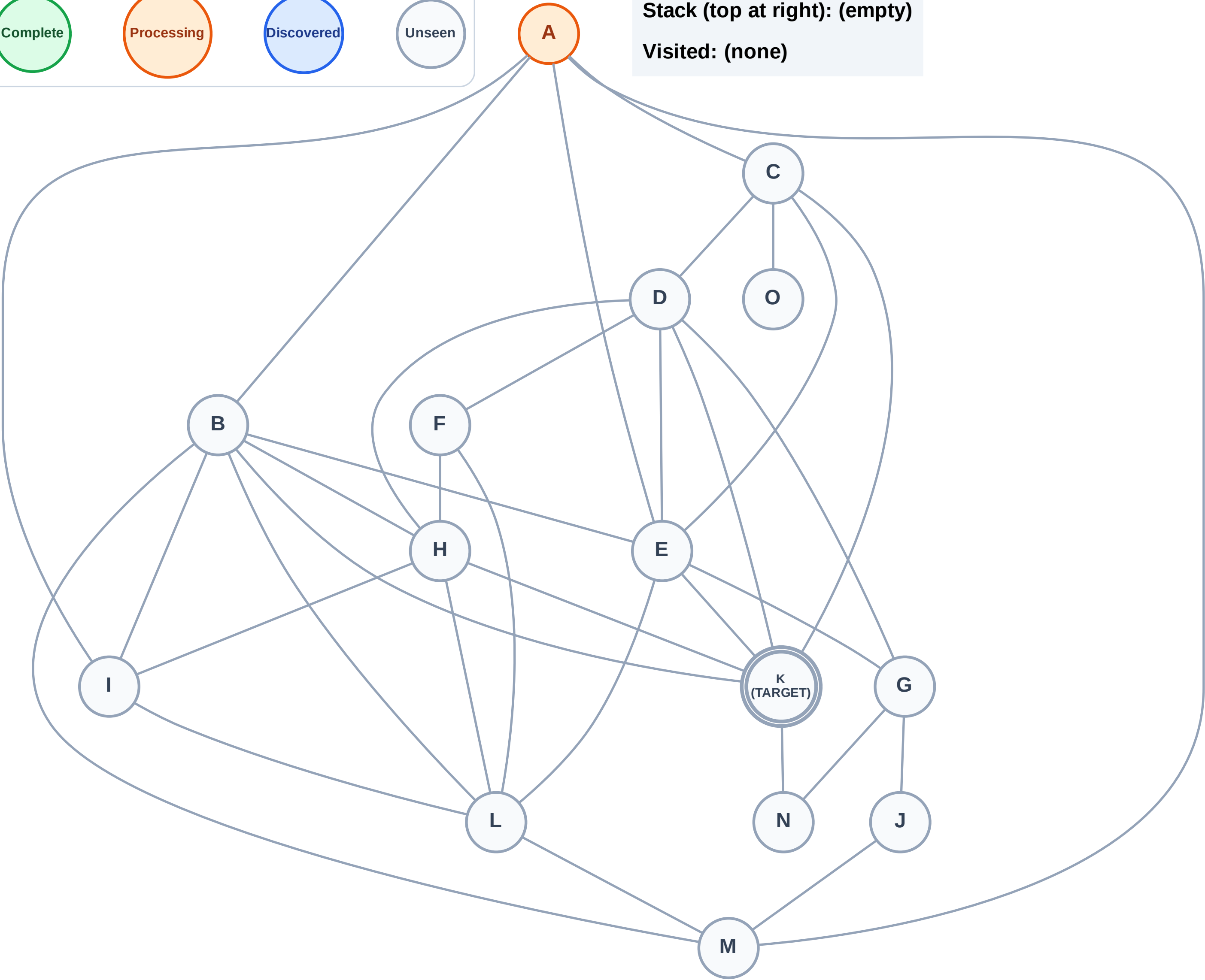
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover M, I, E, C, B from A

Legend

Complete

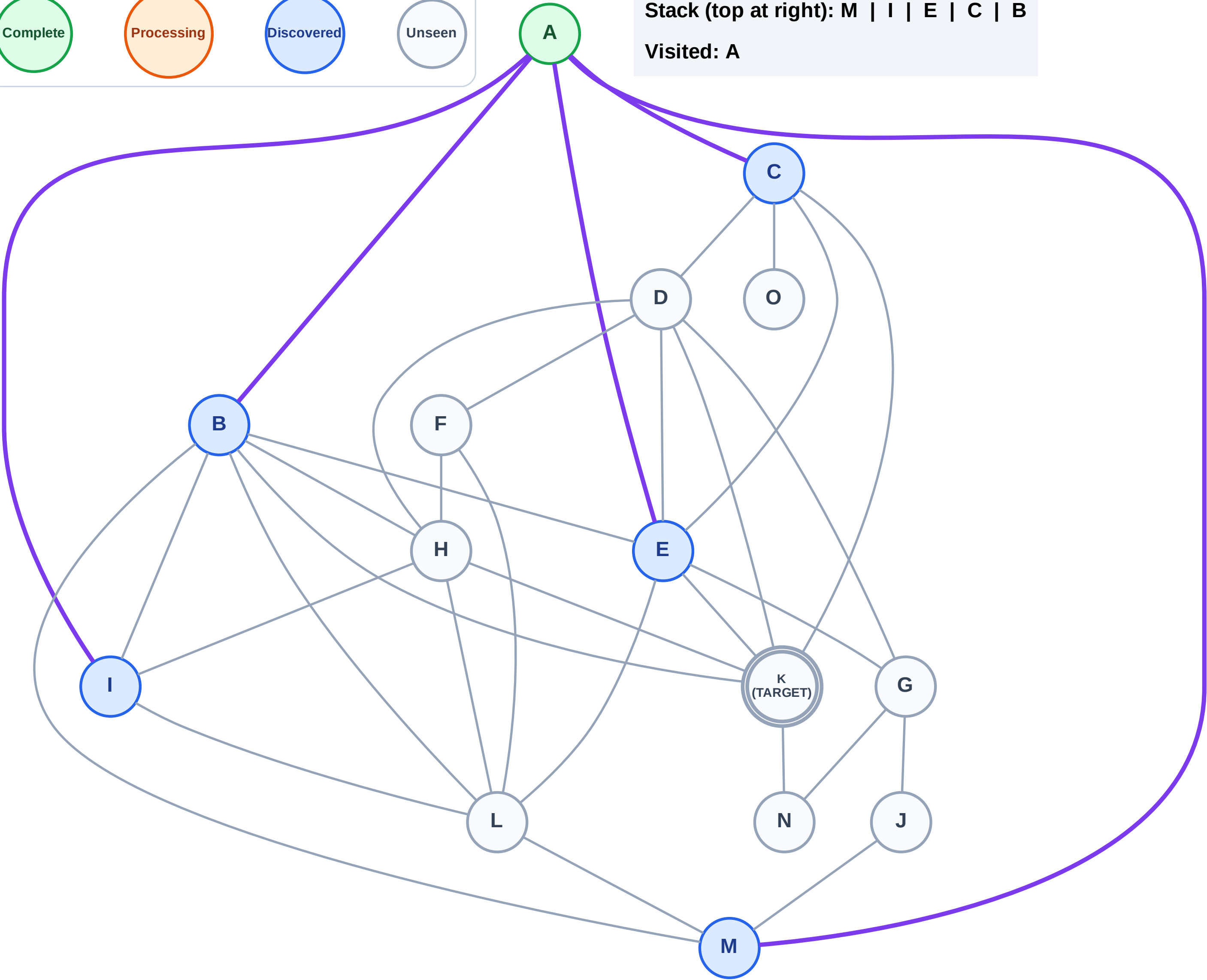
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

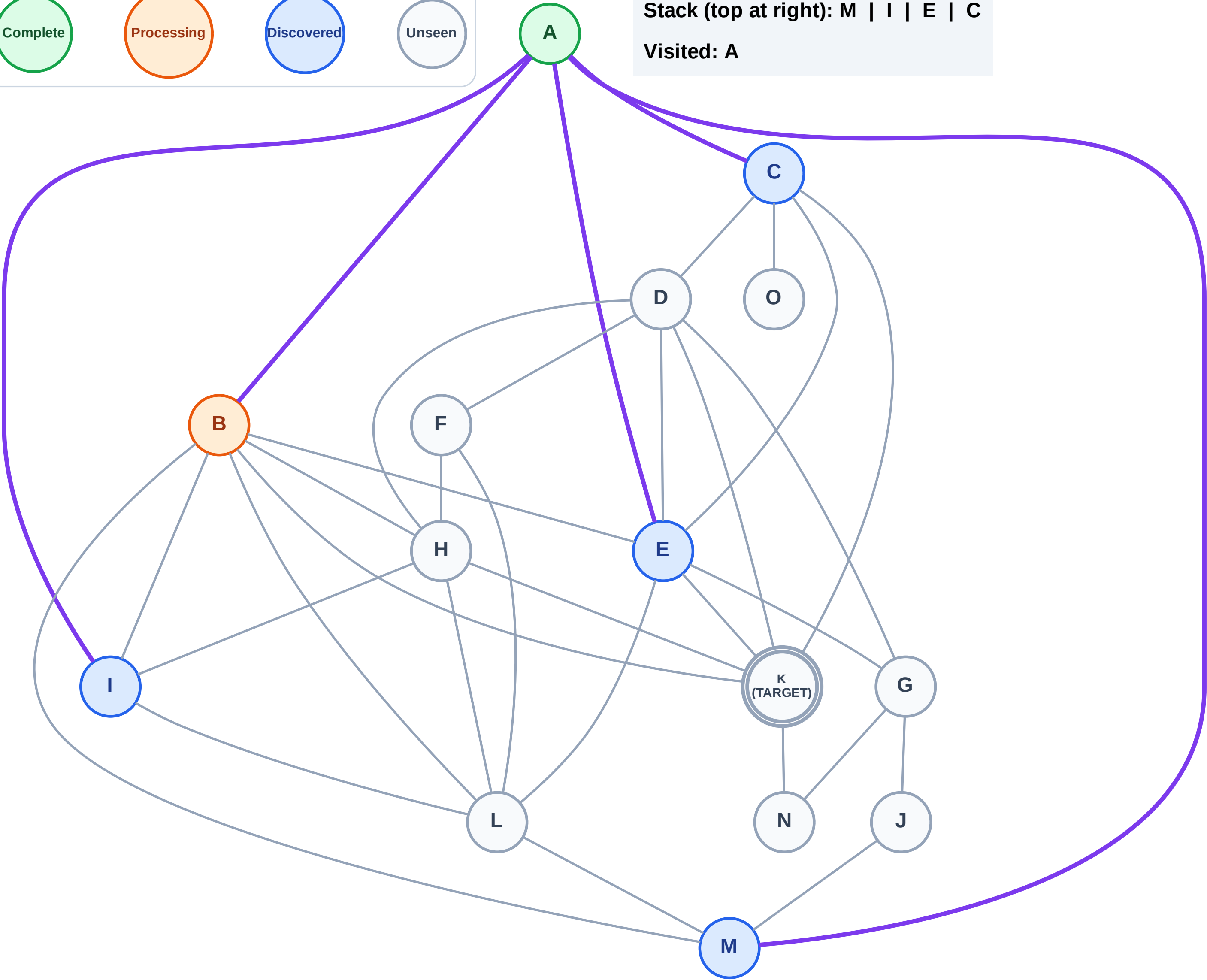
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C

Visited: A



DFS Traversal

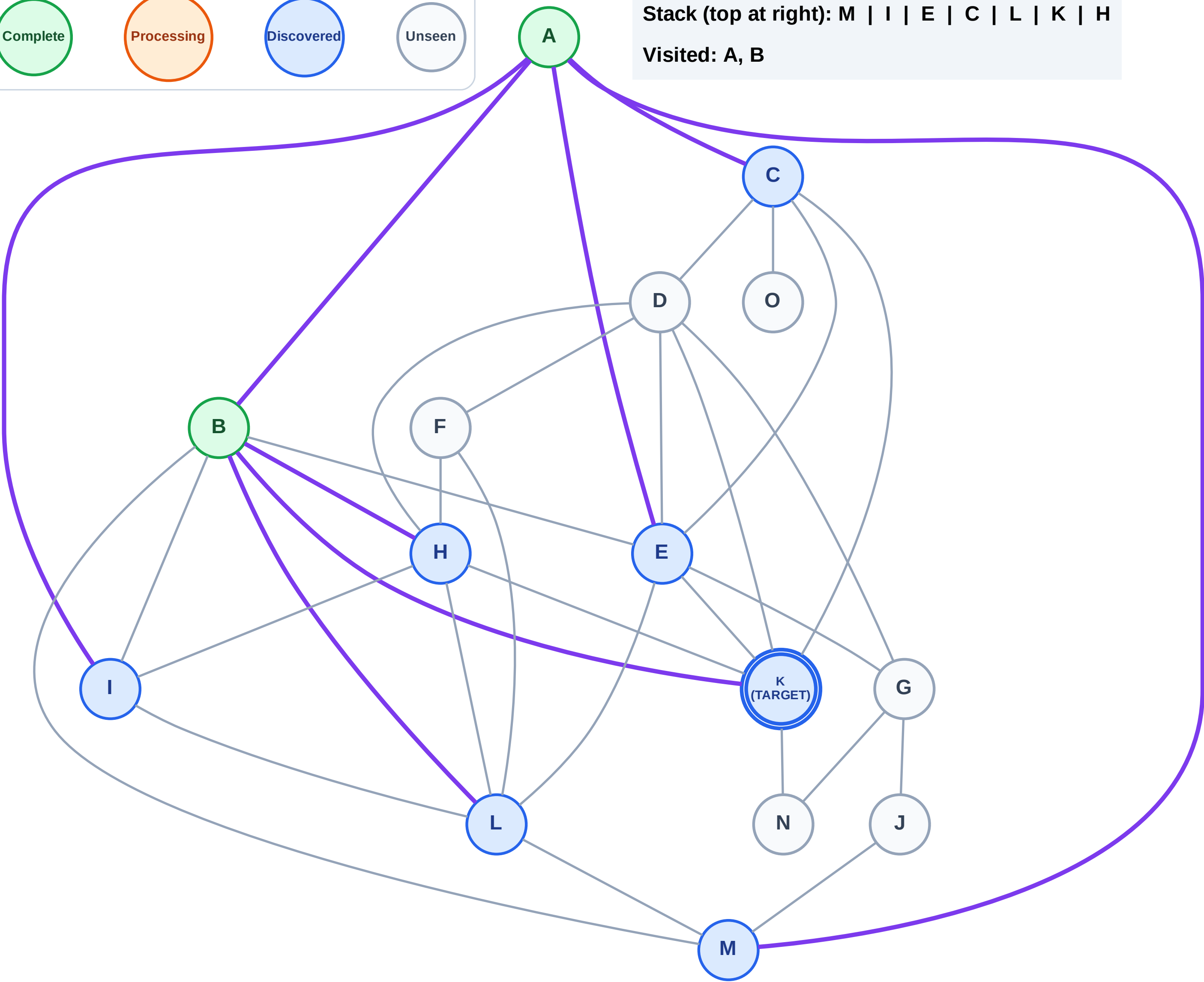
Step 5: Discover L, K, H from B

Legend



Stack (top at right): M | I | E | C | L | K | H

Visited: A, B



DFS Traversal

Step 6: Process node H

Legend

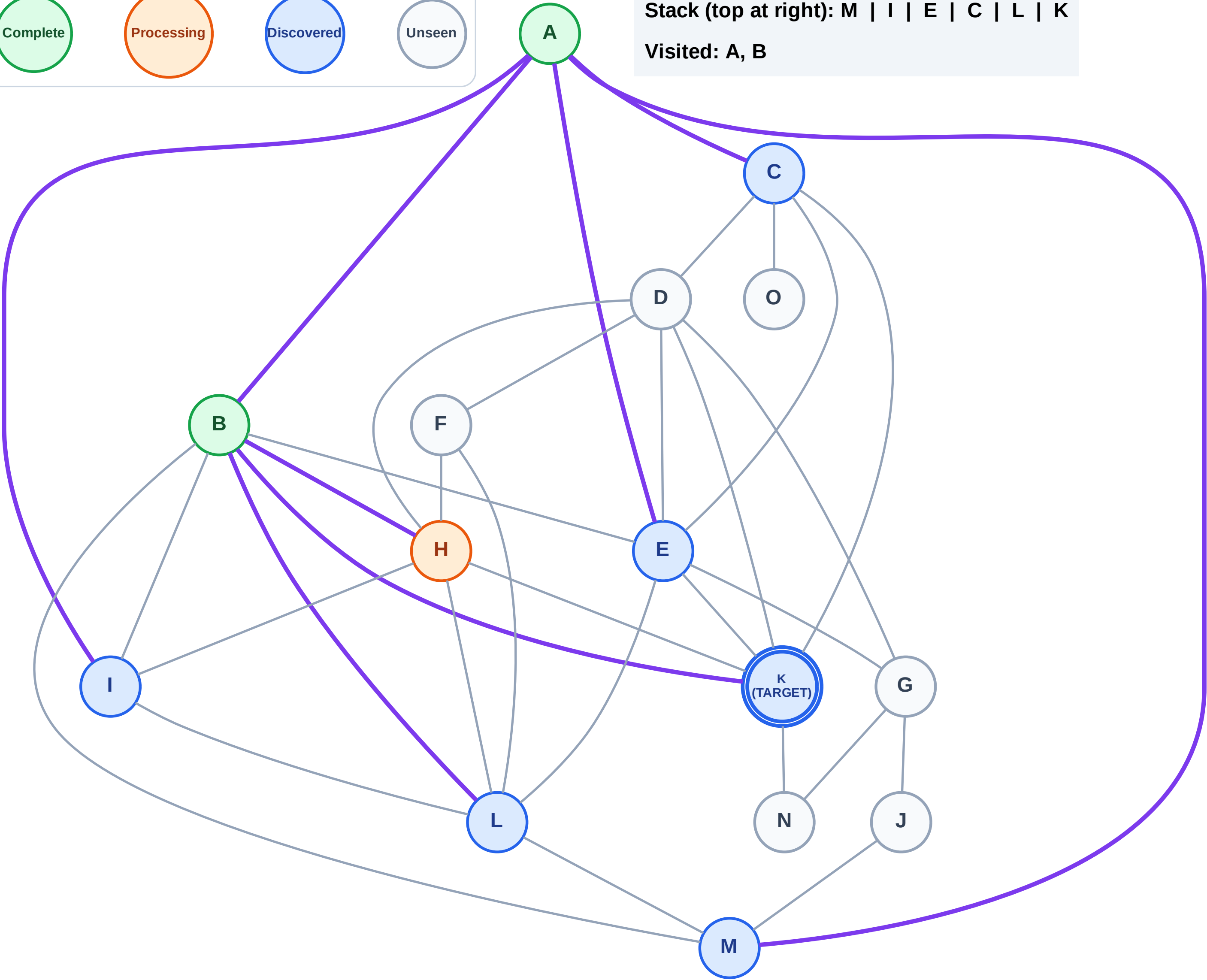
Complete

Processing

Discovered

Unseen

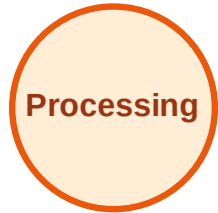
Stack (top at right): M | I | E | C | L | K
Visited: A, B



DFS Traversal

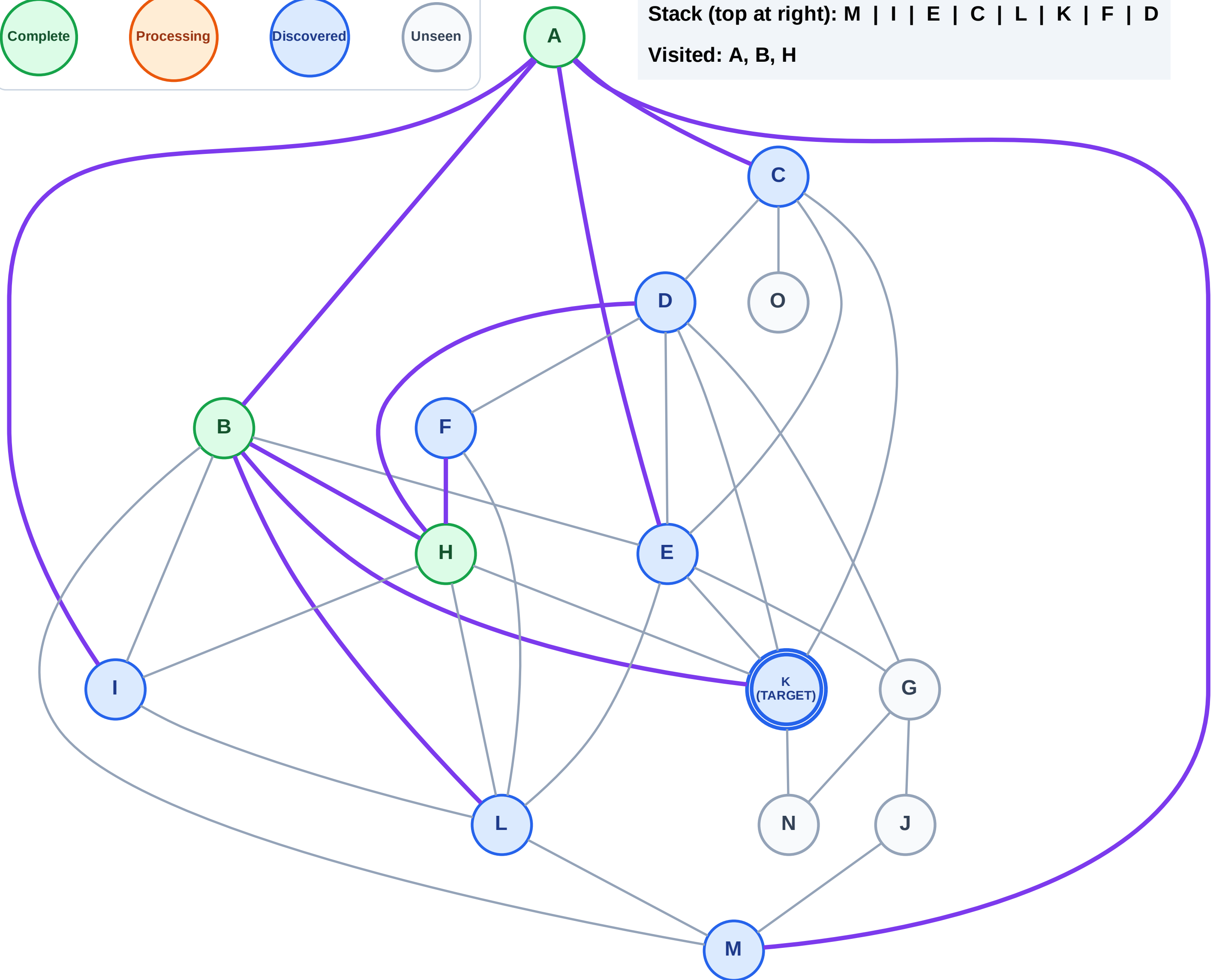
Step 7: Discover F, D from H

Legend



Stack (top at right): M | I | E | C | L | K | F | D

Visited: A, B, H



DFS Traversal

Step 8: Process node D

Legend

Complete

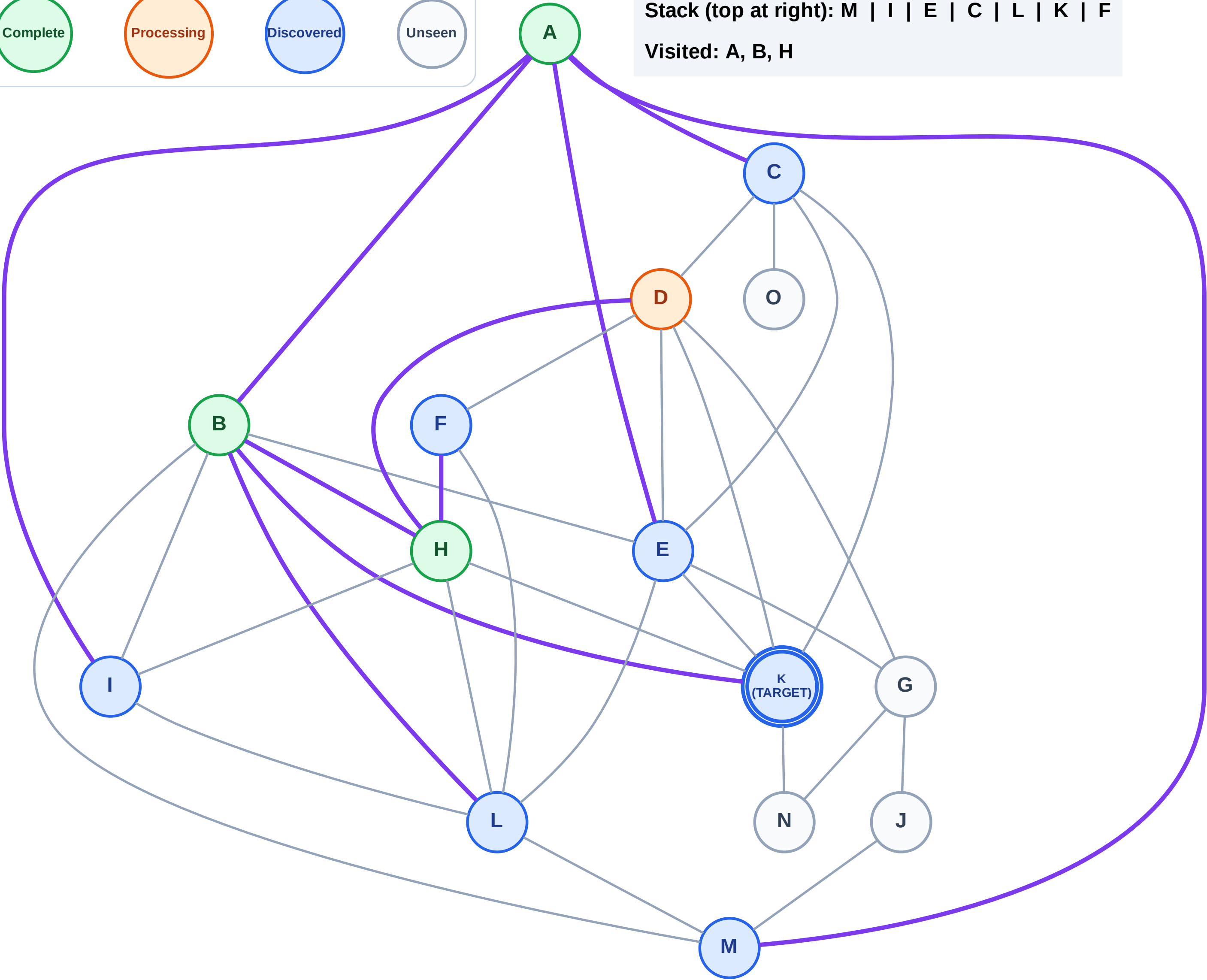
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | L | K | F

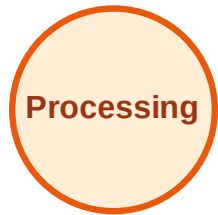
Visited: A, B, H



DFS Traversal

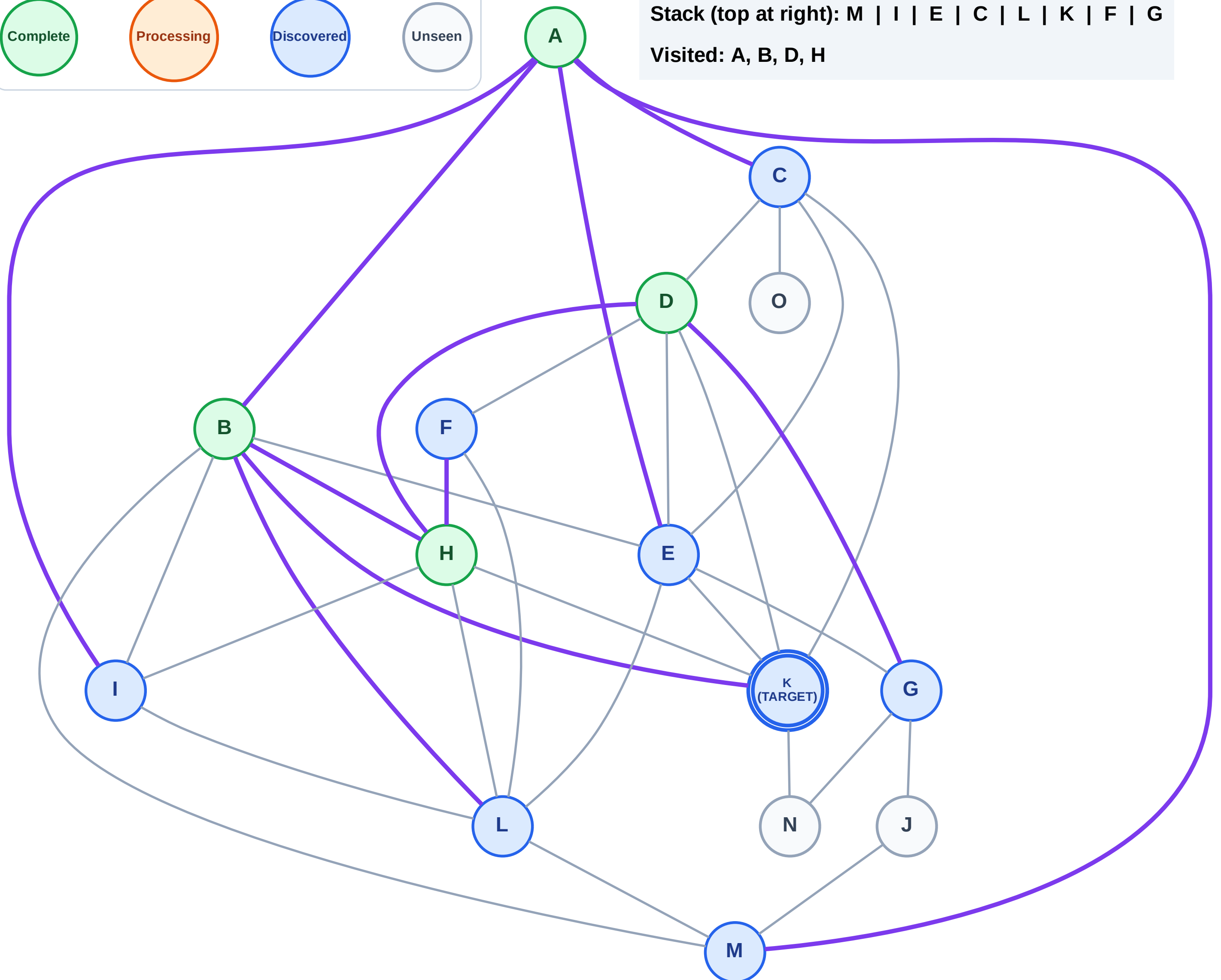
Step 9: Discover G from D

Legend



Stack (top at right): M | I | E | C | L | K | F | G

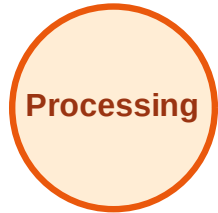
Visited: A, B, D, H



DFS Traversal

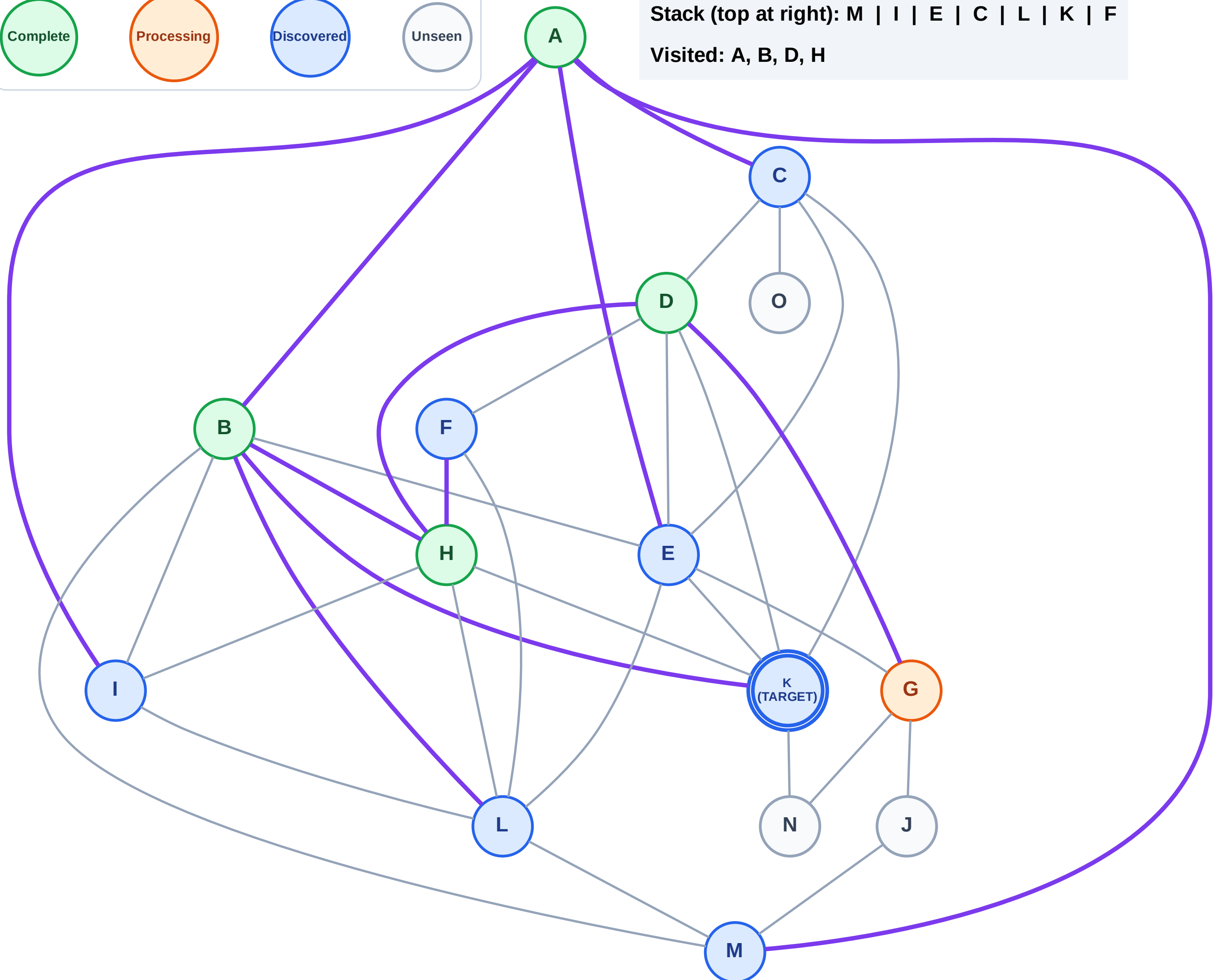
Step 10: Process node G

Legend



Stack (top at right): M | I | E | C | L | K | F

Visited: A, B, D, H



DFS Traversal

Step 11: Discover N, J from G

Legend

Complete

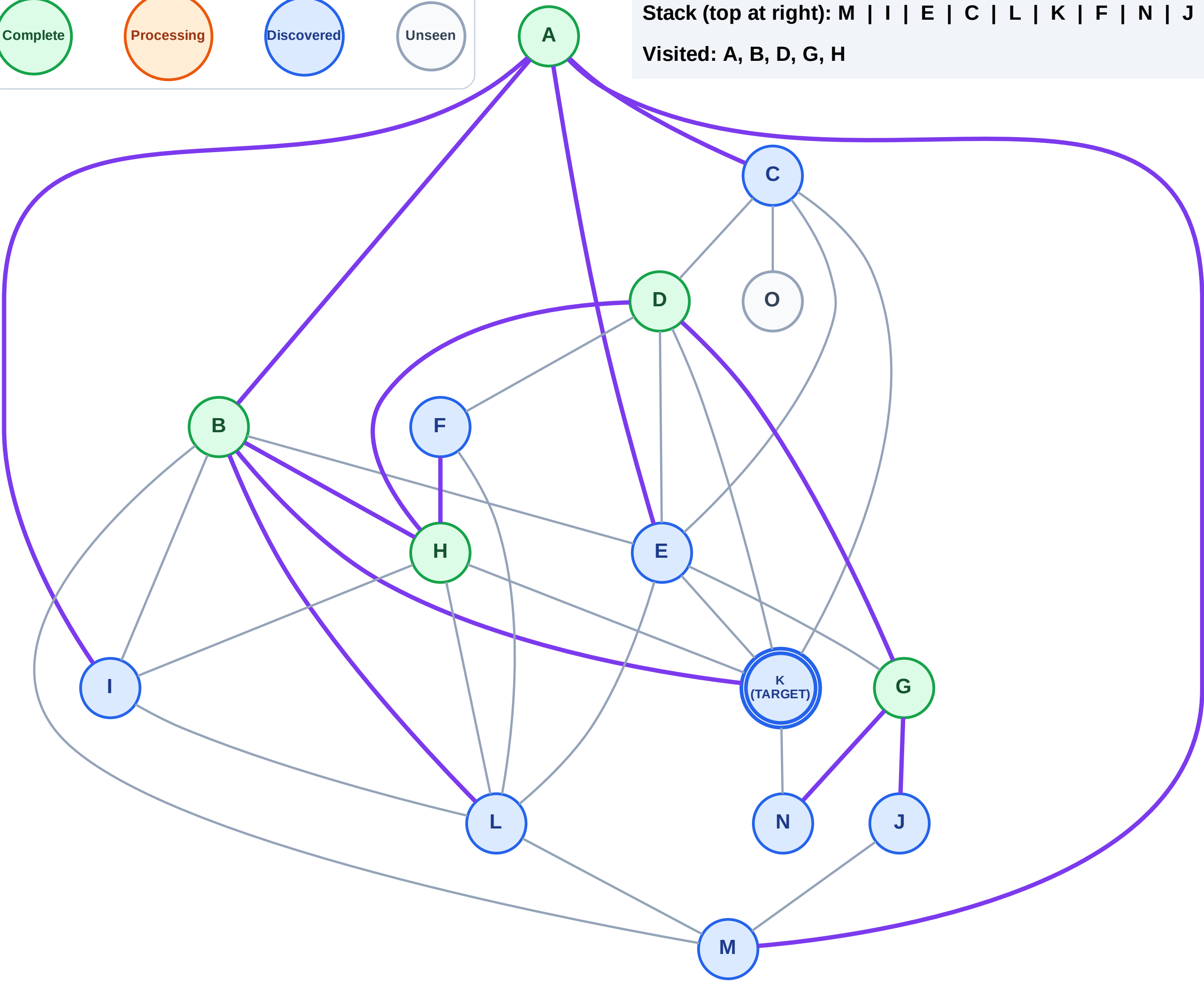
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | L | K | F | N | J

Visited: A, B, D, G, H



DFS Traversal

Step 12: Process node J

Legend

Complete

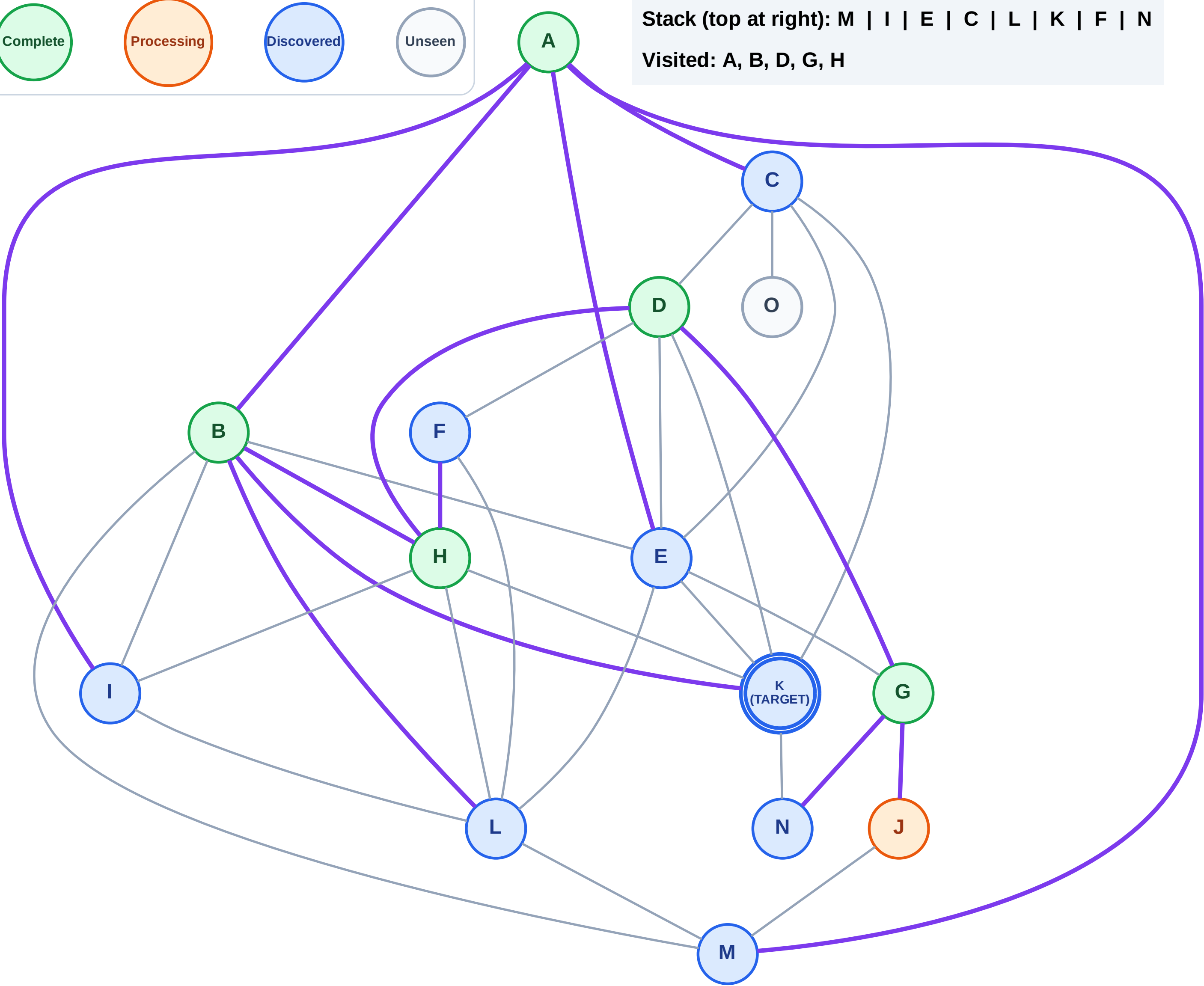
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | L | K | F | N

Visited: A, B, D, G, H



DFS Traversal

Step 13: No new neighbors from J

Legend

Complete

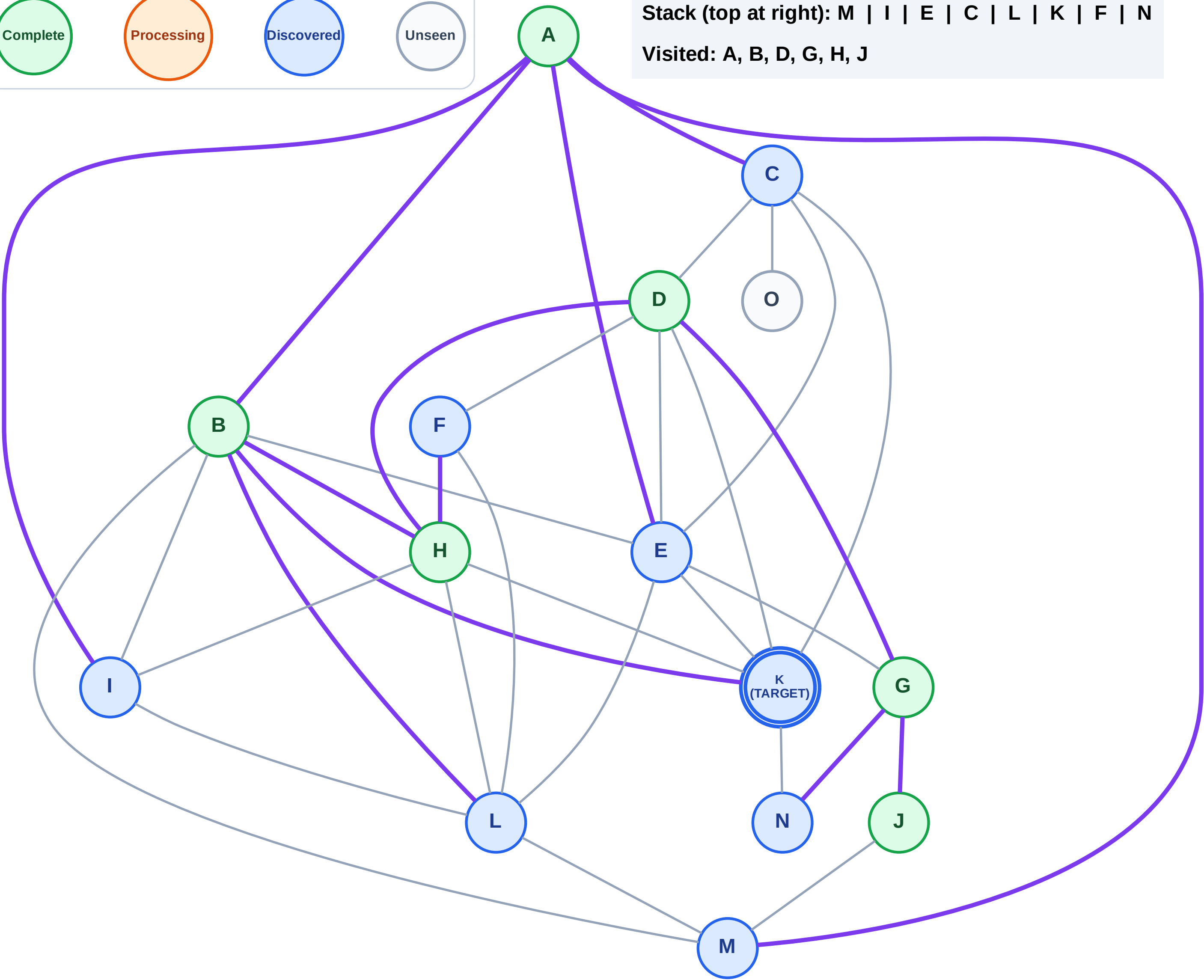
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | L | K | F | N

Visited: A, B, D, G, H, J



DFS Traversal

Step 14: Process node N

Legend

Complete

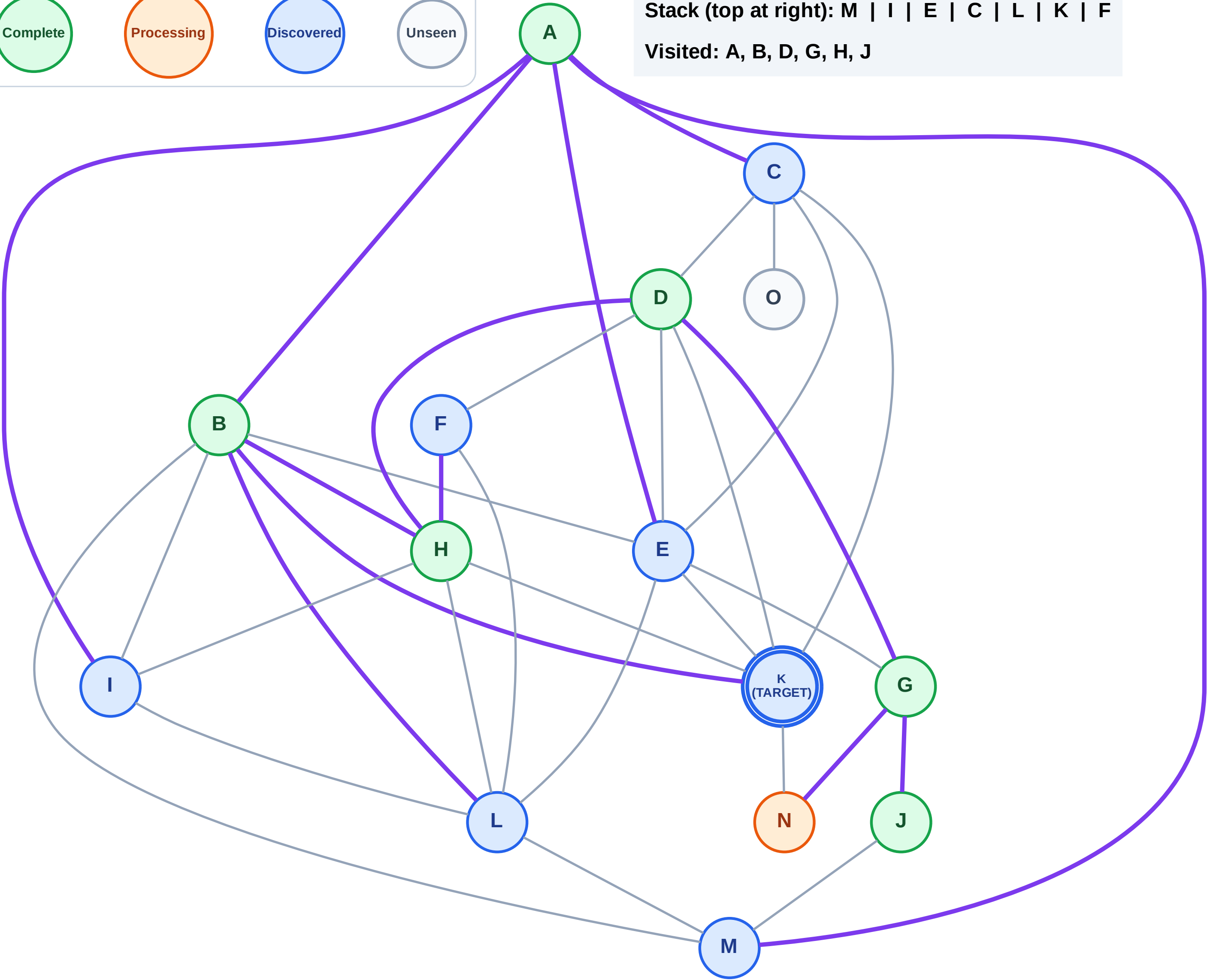
Processing

Discovered

Unseen

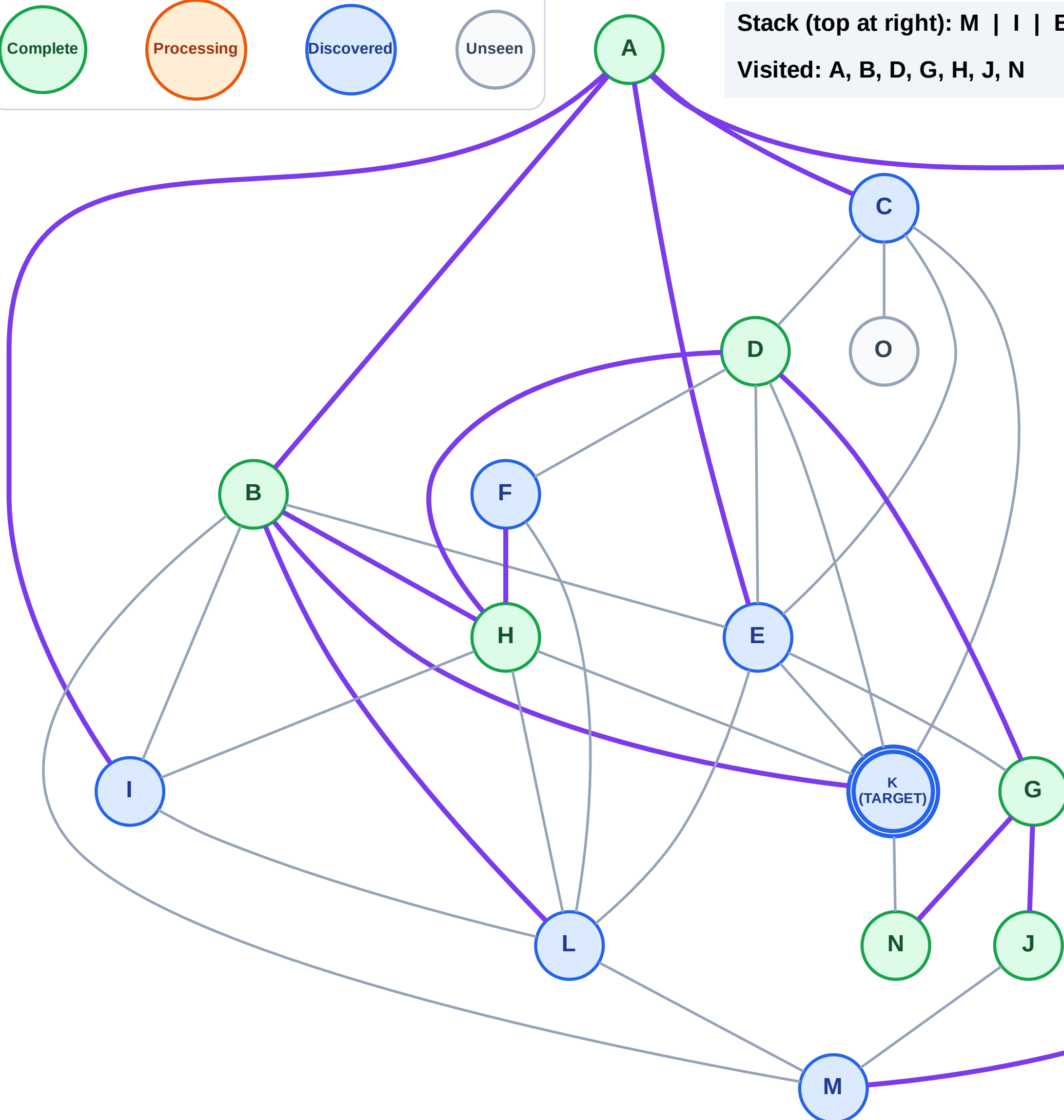
Stack (top at right): M | I | E | C | L | K | F

Visited: A, B, D, G, H, J



Step 15: No new neighbors from N

Step 15: No new neighbors from N



DFS Traversal

Step 16: Process node F

Legend

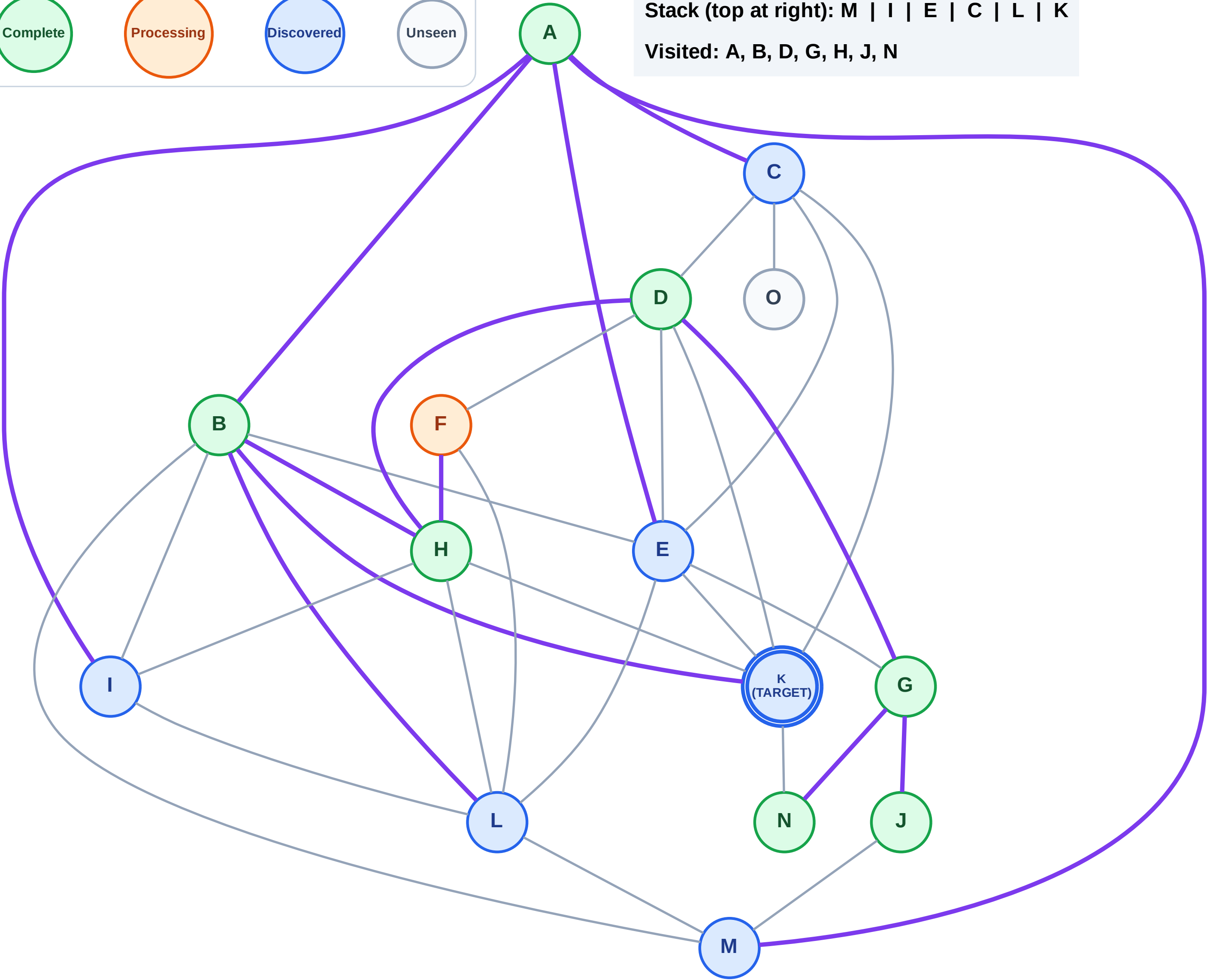
Complete

Processing

Discovered

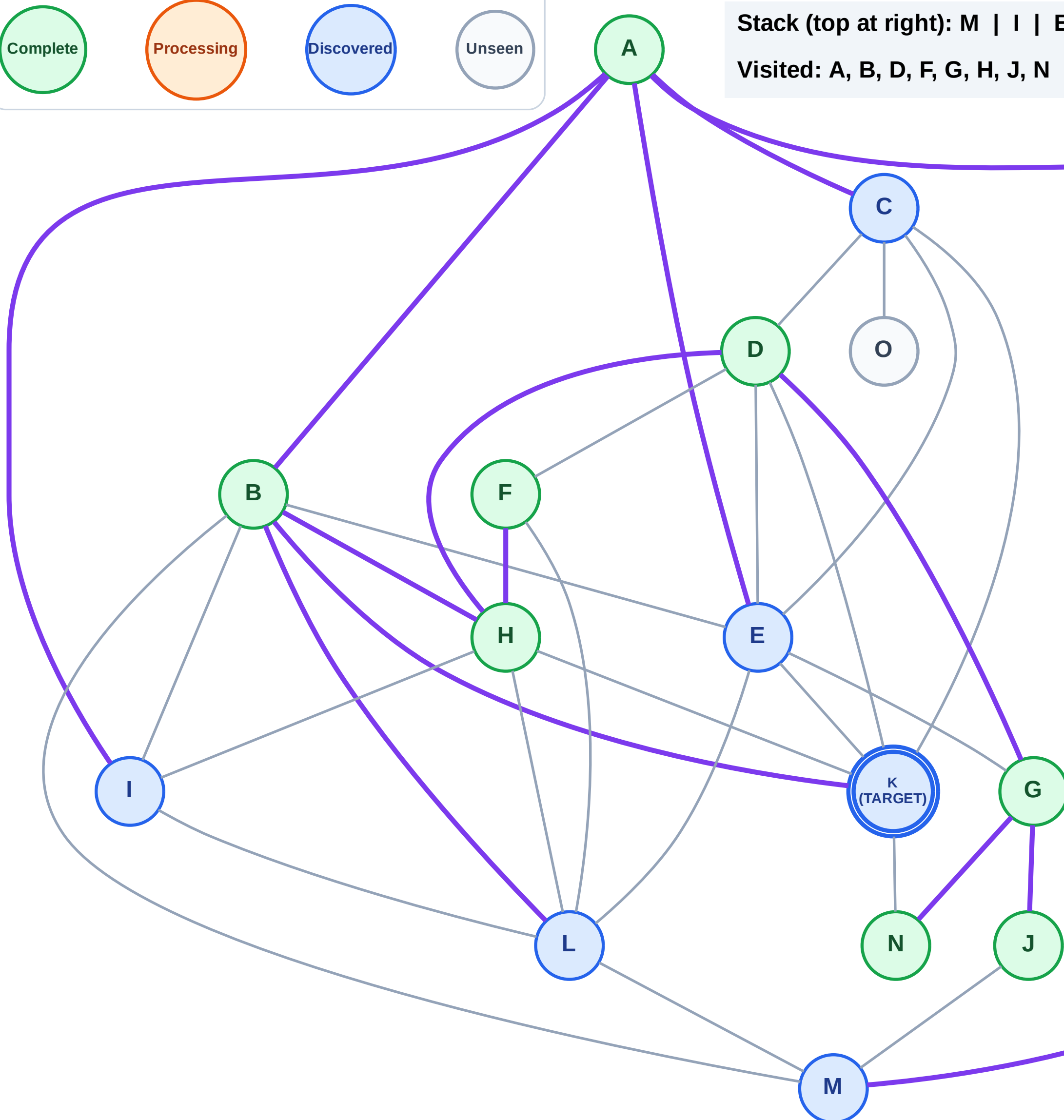
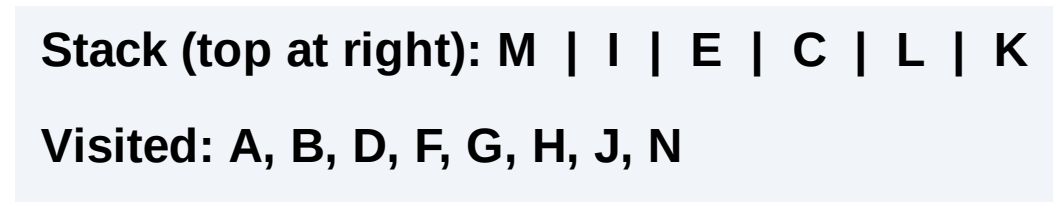
Unseen

Stack (top at right): M | I | E | C | L | K
Visited: A, B, D, G, H, J, N



Step 17: No new neighbors from F

Step 17: No new neighbors from F



DFS Traversal

Step 18: Process node K

Legend

Complete

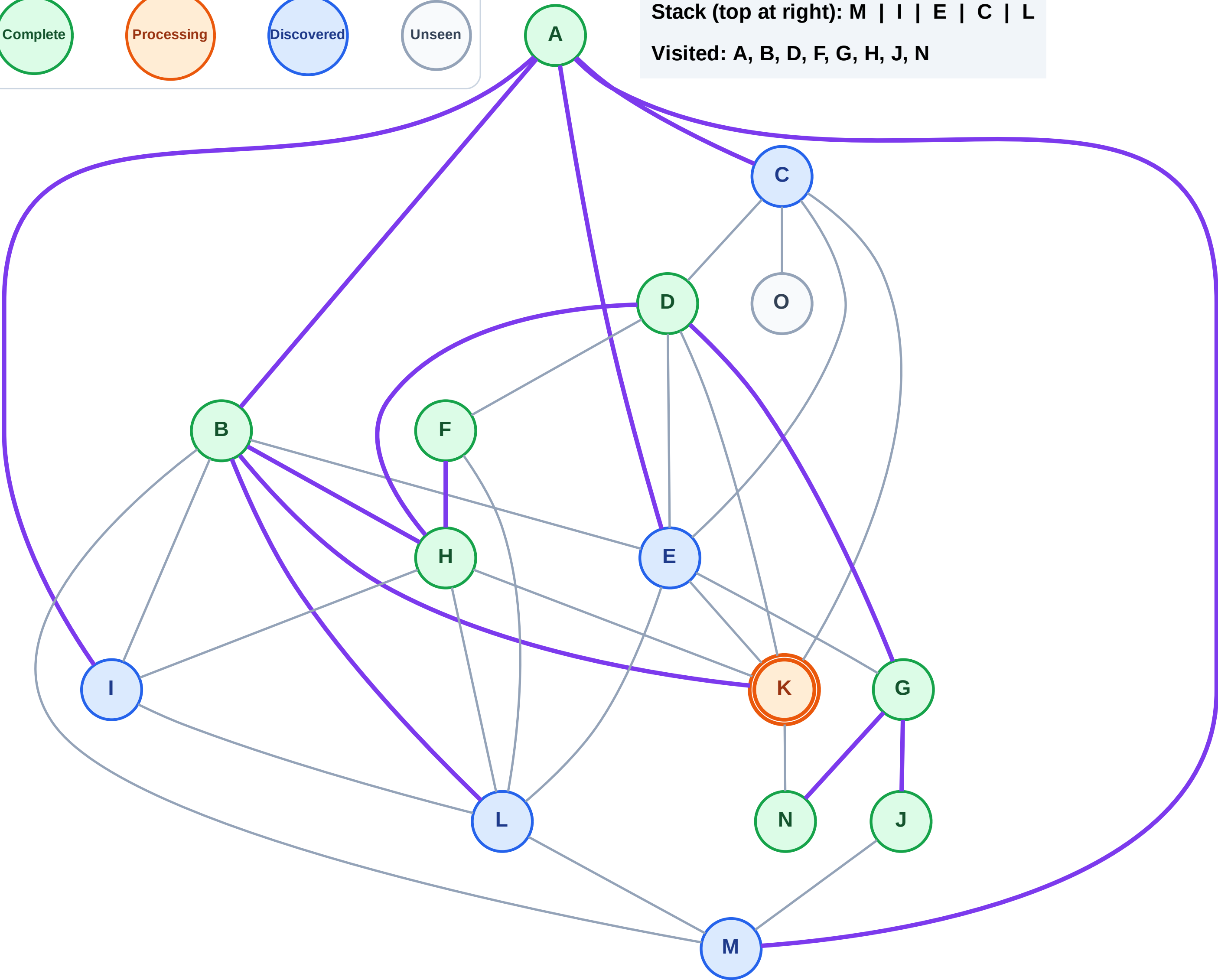
Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | L

Visited: A, B, D, F, G, H, J, N



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | I | E | C | L

Visited: A, B, D, F, G, H, J, K, N

